

Nested Sampling for General Bayesian Computation

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Abstract. Nested sampling estimates directly how the likelihood function relates to prior mass. The evidence (alternatively the marginal likelihood, marginal density of the data, or the prior predictive) is immediately obtained by summation. It is the prime result of the computation, and is accompanied by an estimate of numerical uncertainty. Samples from the posterior distribution are an optional by-product, obtainable for any temperature. The method relies on sampling within a hard constraint on likelihood value, as opposed to the softened likelihood of annealing methods. Progress depends only on the shape of the “nested” contours of likelihood, and not on the likelihood values. This invariance (over monotonic relabelling) allows the method to deal with a class of phase-change problems which effectively defeat thermal annealing.

Keywords: Bayesian computation, evidence, marginal likelihood, algorithm, nest, annealing, phase change, model selection

1 Introduction

Our primary task is to evaluate

$$Z = \text{evidence} = \int L \, dX, \tag{1}$$

where $L = L(\theta)$ is the likelihood function, $dX = \pi(\theta)d\theta$ is the element of prior mass, and θ represents the unknown parameter(s). The probabilistic context of this is usually in the form of Bayes’ theorem, being the product law under background model assumptions $\mathcal{I}$:

$$\begin{aligned} \Pr(D \mid \theta, \mathcal{I}) \times \Pr(\theta \mid \mathcal{I}) &= \Pr(D \mid \mathcal{I}) \times \Pr(\theta \mid D, \mathcal{I}) \\ \text{Likelihood} \times \text{Prior} &= \text{Evidence} \times \text{Posterior} \\ L(\theta) \times \pi(\theta)d\theta &= Z \times p(\theta)d\theta. \end{aligned} \tag{2}$$

Here D are the acquired data which let us modulate our prior belief $dX = \pi(\theta)d\theta$ into posterior $dP = p(\theta)d\theta$. Prior and posterior, as always, are normalised to unit total. Inputs L and π yield outputs Z and p .

Calculating the value of Z allows different model assumptions to be compared through the ratios of evidence values known as Bayes factors. Presenting Z thus lets the results be future-proof, in that future models can be compared with the current one, without having to re-do the current calculation. Giving the value of Z is a courtesy to other workers who may wish to perform model selection, and ought to be a standard part of rational enquiry. It is one half of the output from a Bayesian calculation, the other half being the posterior. Oddly,

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嵌套采样：通用贝叶斯计算

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摘要。嵌套采样直接估计似然函数与先验质量的关系。证据（或称为边缘似然、数据边缘密度或先验预测）通过求和立即获得。这是计算的主要结果，并伴随有数值不确定性的估计。后验分布的样本是可选的副产品，可通过任何温度获得。该方法依赖于在似然值上的硬约束内进行采样，而不是退火方法中的软化似然。进展仅取决于似然“嵌套”轮廓的形状，而与似然值无关。这种不变性（在单调重标下）允许该方法处理一类相位变化问题，这些问题实际上使热退火失效。

关键词：贝叶斯计算、证据、边缘似然、算法、嵌套、退火、相位变化、模型选择

1 简介

我们的主要任务是评估

$$Z = \text{evidence} = \int L \, dX, \tag{1}$$

其中 $L = L(\theta)$ 是似然函数， $dX = \pi(\theta)d\theta$ 是先验质量的元素， θ 表示未知参数。这种概率背景通常以贝叶斯定理的形式出现，在背景模型假设 $\mathcal{I}$ 下是乘法律律：

$$\begin{aligned} \Pr(D \mid \theta, \mathcal{I}) \times \Pr(\theta \mid \mathcal{I}) &= \Pr(D \mid \mathcal{I}) \times \Pr(\theta \mid D, \mathcal{I}) \\ \text{Likelihood} \times \text{Prior} &= \text{Evidence} \times \text{Posterior} \\ L(\theta) \times \pi(\theta)d\theta &= Z \times p(\theta)d\theta. \end{aligned} \tag{2}$$

这里 D 是我们获得的数据，让我们将先验信念 $dX = \pi(\theta)d\theta$ 调整为后验 $dP = p(\theta)d\theta$ 。像往常一样，先验和后验都归一化为单位总和。输入 L 和 π 产生输出 Z 和 p 。

计算 Z 的值允许通过证据值比率（即贝叶斯因子）比较不同的模型假设。展示 Z 因此能让结果具备前瞻性，即未来的模型可以与当前模型进行比较，而无需重新计算当前结果。给出 Z 的值是对其他可能希望进行模型选择的工作者的礼貌，也应当是理性探究的标准部分。这是贝叶斯计算输出的一半，另一半是后验概率。奇怪的是，

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the quantity currently lacks a universal name, and the author favours the crisp single word “evidence”, which is growing in popularity (MacKay (2003)) and for which no other standard technical definition has yet been agreed. “Marginal likelihood” describes how it is (usually but not always) constructed. “Prior predictive” describes how it is (usually but not always) used. “Evidence” denotes what it is.

Historically, following Metropolis et al. (1953) and Hastings (1970), algorithms such as Markov chain Monte Carlo (MCMC) have been principally designed for the posterior. Indeed, standard MCMC yields only a set of samples representing the normalised posterior, and fails to give the evidence at all. Obtaining the evidence has required considerable extra work, usually involving a sequence of intermediate distributions that bridge between prior and posterior, as in thermodynamic integration reviewed with generalisations by Gelman and Meng (1998). This is unfortunate, because the extra computational difficulty and the lack of standard terminology suggest that the evidence value is an optional by-product, rather than a quantity of central importance. Nested sampling reverses the historical approach. The evidence is now the prime target, with representative posterior samples available as the optional by-product.

The paper starts with the idea of sorting points θ by their likelihood values, which are then summed to give the evidence. Of course, there are usually far too many points to do this explicitly, so nested sampling simulates the operation statistically. The evidence then becomes accompanied by a corresponding numerical uncertainty. A methodological section argues that nested sampling is Bayesian in nature. With the basic method in place, it is possible to estimate the density of states, to obtain samples from the posterior, and to quantify arbitrary properties of θ . These sections complete the formal development. Nested sampling is then compared with the conventional approach of annealing, and is shown by examples and limiting cases to be wider in scope. The paper concludes with an overview, and an Appendix with a simple ‘C’ program.

2 Sorting

The evaluation of $\int L dX$ looks like a straightforward problem of numerical analysis. Simplistically, one might raster over underlying coordinates θ to evaluate $\int L(\theta)\pi(\theta)d\theta$. However, this rapidly becomes impractical as soon as θ has more than a very few dimensions. Instead, we will use the prior X directly.

Prior mass X can be accumulated from its elements dX in any order, so define

$$X(\lambda) = \int_{L(\theta) > \lambda} \pi(\theta) d\theta \quad (3)$$

as the cumulant prior mass covering all likelihood values greater than λ . As λ increases, the enclosed mass X decreases from 1 to 0. Writing the inverse function as $L(X)$, *i.e.* $L(X(\lambda)) \equiv \lambda$, the evidence becomes a one-dimensional integral over unit range

$$Z = \int_0^1 L(X) dX \quad (4)$$

in which the integrand is positive and decreasing (Figure 1), so it has to be well behaved. Accomplishing this transformation from θ to X involves dividing the unit prior mass into tiny elements, and sorting them by likelihood.

这个量目前还没有一个通用的名称，作者更倾向于使用清晰的单个词 “\evidence\”，这个词正日益流行（MacKay (2003)），并且目前还没有其他标准的技术定义。边缘似然描述了它的构建方式（通常但并非总是如此）。先验预测描述了它的使用方式（通常但并非总是如此）。证据指的是它本身。

历史上，遵循Metropolis等人（1953年）和Hastings（1970年）的研究，马尔可夫链蒙特卡洛（MCMC）等算法主要针对后验分布进行设计。事实上，标准MCMC仅产生一组代表归一化后验分布的样本，而完全无法给出证据。获取证据需要大量额外工作，通常涉及一系列连接先验分布和后验分布的中间分布，例如Gelman和Meng（1998年）综述的热力学积分及其推广。这很不幸，因为额外的计算难度和缺乏标准术语表明，证据值是一个可选的副产品，而不是一个核心重要的量。嵌套采样颠覆了历史方法。证据现在成为主要目标，而代表性后验样本则作为可选副产品提供。

本文从按似然值排序点的想法开始，这些值随后被求和以给出证据。当然，通常有太多点无法显式进行此操作，因此嵌套采样进行统计模拟。证据随后伴随着相应的数值不确定性。方法论部分论证了嵌套采样的贝叶斯性质。基本方法确定后，可以估计态密度，从后验分布中获取样本，并量化 θ 的任意属性。这些部分完成了形式化发展。然后，嵌套采样与退火的传统方法进行比较，并通过示例和极限情况证明其适用范围更广。本文最后概述了内容，并附有一个简单的 ‘C’ 程序。

2 排序

$\int L dX$ 的评估看似是一个简单的数值分析问题。简单来说，人们可能会在底层坐标 θ 上扫描以评估 $\int L(\theta)\pi(\theta)d\theta$ 。然而，一旦 θ 的维度超过几个，这就会变得不切实际。相反，我们将直接使用先验 X 。

先验质量 X 可以从其元素 dX 以任何顺序累积，因此定义

$$X(\lambda) = \int_{L(\theta) > \lambda} \pi(\theta) d\theta \quad (3)$$

为覆盖所有大于 λ 的似然值的累积先验质量。随着 λ 的增加，包含的质量 X 从1减少到0。将逆函数写为 $L(X)$ ，*i.e.* $L(X(\lambda)) \equiv \lambda$ ，证据变成一个单位范围内的单变量积分

$$Z = \int_0^1 L(X) dX \quad (4)$$

在积分函数为正且递减的情况下（图1），它必须表现良好。完成从 θ 到 X 的这种转换，需要将单位先验质量分割成微小元素，并按可能性进行排序。

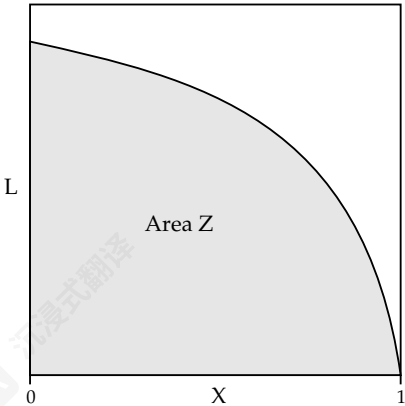


Figure 1: Likelihood function with area Z.

A very simple example, on a 4×4 grid of two-dimensional θ , is shown in the table (5) of likelihood values ascribed to its 16 cells of equal prior mass $\frac{1}{16}$.

0	8	15	3
11	24	22	10
19	30	26	16
9	23	18	6

$L =$

(5)

Our plan is to proceed as if we could sort these elements by likelihood, in the above example to $L = (30,26,24,23,22,19,18,16,15,11,10,9,8,6,3,0)$, whence

$$Z = \frac{30}{16} + \frac{26}{16} + \frac{24}{16} + \frac{23}{16} + \frac{22}{16} + \frac{19}{16} + \frac{18}{16} + \frac{16}{16} + \frac{15}{16} + \frac{11}{16} + \frac{10}{16} + \frac{9}{16} + \frac{8}{16} + \frac{6}{16} + \frac{3}{16} + \frac{0}{16} = 15,$$

to be evaluated right-to-left into domains of progressively greater likelihood. The likelihood corresponding to (say) $X = \frac{1}{5}$, being one fifth of the way along the sequence so falling into the fourth cell out of sixteen, is $L(X=0.2) = 23$.

As a technicality we may need to resolve ties between points of equal L . A point k , which has coordinates θ_k and corresponding likelihood $L_k = L(\theta_k)$, can also be assigned a label ℓ_k , chosen from some library large enough that repeats are not expected. Random samples from **Uniform**(0,1) suffice, as would a cryptographic identification key derived from θ , or almost anything else. Labels parameterise within each likelihood contour, and extend the likelihood to

$$L_k^+ = L_k + \epsilon \ell_k,$$

(6)

where ϵ is some tiny coefficient that never affects numerical likelihood values (which are always held to finite precision), but nevertheless enables an unambiguous ranking of the points, even where raw likelihoods are equal. For clarity, we mostly ignore this refinement hereafter.

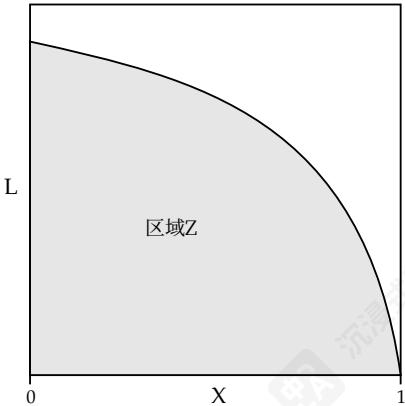


图1: 似然函数与区域Z。

一个非常简单的例子，在一个二维 θ 的 4×4 网格上，如表（5）所示，它将等先验质量为 1/16的16个单元格分配了可能性值。

0	8	15	3
11	24	22	10
19	30	26	16
9	23	18	6

$L =$

(5)

我们的计划是假设我们可以根据似然对这些元素进行排序，在上述示例中为 $L = (30,26,24,23,22,19,18,16,15,11,10,9,8,6,3,0)$ ，从而

$$Z = \frac{30}{16} + \frac{26}{16} + \frac{24}{16} + \frac{23}{16} + \frac{22}{16} + \frac{19}{16} + \frac{18}{16} + \frac{16}{16} + \frac{15}{16} + \frac{11}{16} + \frac{10}{16} + \frac{9}{16} + \frac{8}{16} + \frac{6}{16} + \frac{3}{16} + \frac{0}{16} = 15,$$

从右到左评估到似然逐渐增大的域。对应于（例如） $X = \frac{1}{5}$ 的似然，由于它位于序列的 6分之一处，落入十六个单元格中的第四个单元格，因此是 $L(X=0.2) = 23$ 。

作为一个技术性细节，我们可能需要解决在具有相等 L 的点之间的平局。一个具有坐标 θ_k 和相应似然 $L_k = L(\theta_k)$ 的点 k ，也可以分配一个标签 ℓ_k ，该标签从某个足够大的库中选择，以避免重复。来自**Uniform**(0,1)的随机样本，就像从 θ 派生的加密身份识别密钥，或几乎任何其他东西。标签在每个似然轮廓内参数化，并将似然扩展到

$$L_k^+ = L_k + \epsilon \ell_k,$$

(6)

ϵ 处有一些微小的系数，它永远不会影响数值可能性值（这些值始终以 6有穷精度存储），但仍然能够对点进行无歧义的排序，即使原始可能性值相等。为清晰起见，我们在此之后大多忽略这一6注记。

3 Integration

Coordinate-dependent complications of geometry, topology, even dimensionality, are all annihilated by the sorting operation, and the remaining task is easy and well understood. Suppose that we knew how to evaluate the likelihood as $L_i = L(X_i)$ at a right-to-left sequence of m points

$$0 < X_m < \cdots < X_2 < X_1 < 1. \quad (7)$$

Any convenient numerical recipe would then estimate Z as a weighted sum

$$Z \leftarrow \sum_{i=1}^m w_i L_i \quad (8)$$

of these values, in which the area in Figure 1 is approximated as a set of columns of height L and width $w \sim \Delta X$.

Because $L(X)$ is non-increasing, it is bounded below by any value evaluated at larger X . Hence $w_i = X_i - X_{i+1}$ with $X_{m+1} = 0$ gives a lower bound

$$Z = \int_0^1 L dX \geq \sum_{i=1}^m (X_i - X_{i+1}) L_i. \quad (9)$$

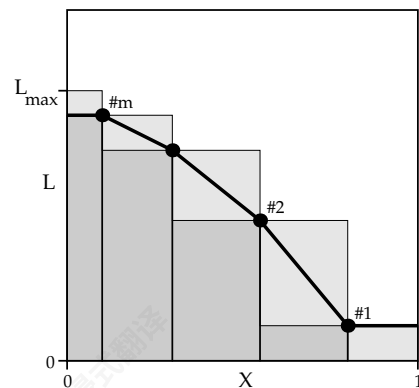


Figure 2: Lower bound (dark shading) and upper bound (all shading) on area. The thick line indicates the trapezoidal rule.

There is a similar upper bound (Figure 2) from $w_i = X_{i-1} - X_i$ with $X_0 = 1$,

$$Z = \int_0^1 L dX \leq \sum_{i=1}^m (X_{i-1} - X_i) L_i + X_m L_{\max}, \quad (10)$$

where $L_{\max}$ is the maximum likelihood value to be found as $X \rightarrow 0$. Technically, $L_{\max}$ is not determined by nested (or any other) sampling. There could always remain some tiny volume containing huge and dominant likelihood values, unless that can be ruled out by some global analysis (as when a Gaussian likelihood factor cannot exceed $1/\sqrt{2\pi\sigma}$). However, when judging that a run can be terminated, we implicitly assert that any increase in L beyond the highest

3 积分

与几何、拓扑甚至维度的坐标相关复杂性问题的，都会被排序操作所消除，剩下的任务简单且易于理解。假设我们知道如何以 $L_i = L(X_i)$ 的方式评估 m 点从右到左序列中的可能性

$$0 < X_m < \cdots < X_2 < X_1 < 1. \quad (7)$$

任何方便的数值方法都会将 Z 估计为加权求和

$$Z \leftarrow \sum_{i=1}^m w_i L_i \quad (8)$$

这些值的组合，其中图1中的面积近似为一系列高度为 L 、宽度为 $w \sim \Delta X$ 的柱子。

因为 $L(X)$ 是非递增的，它被任何在较大 X 处评估的值所下界。因此 $w_i = X_i - X_{i+1}$ 与 $X_{m+1} = 0$ 给出下界

$$Z = \int_0^1 L dX \geq \sum_{i=1}^m (X_i - X_{i+1}) L_i. \quad (9)$$

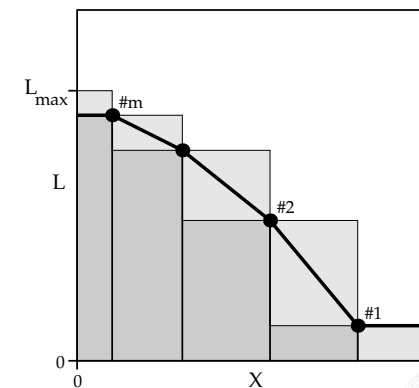


图2：面积的下界（深色阴影）和上界（所有阴影）。粗线表示梯形规则。

存在一个类似的上界（图2）来自 $w_i = X_{i-1} - X_i$ 与 $X_0 = 1$,

$$Z = \int_0^1 L dX \leq \sum_{i=1}^m (X_{i-1} - X_i) L_i + X_m L_{\max}, \quad (10)$$

$L_{\max}$ 在哪里能找到最大似然值 $X \rightarrow 0$ 。技术上讲， $L_{\max}$ 不是由嵌套（或其他任何）采样确定的。总可能存在一些极小的体积包含巨大的主导似然值，除非能通过某种全局分析排除这种情况（例如，高斯似然因子不能超过 $1/\sqrt{2\pi\sigma}$ ）。然而，在判断是否可以终止运行时，我们隐含地断定，任何超出当前最高值的 L 增加都是无关紧要的。在此前提下，上限（10）成立，数值积分误差最多为 $L_{\max} (X \rightarrow 0) L_{\max}$ ，即上限与下限之间的差值。

value yet found is not consequential. With this proviso, the upper limit (10) holds, and errors from numerical integration are at most $\mathcal{O}(N^{-1})$ — that being the difference between upper and lower bounds.

The trapezoidal rule $w_i = \frac{1}{2}(X_{i-1} - X_{i+1})$, with reflecting boundary conditions $X_0 = 2 - X_1$ and $X_{m+1} = -X_m$ to avoid awkward behaviour, reduces this to $\mathcal{O}(N^{-2})$ in most cases. The integrand is already quite well behaved, so we do not expect useful improvement over the trapezoid rule (or similar) denoted by the “ $\longleftarrow$ ” arrow.

4 Evidence

The integral for Z is dominated by wherever the bulk of the posterior mass is to be found. Typically, this occupies a small fraction e^{-H} of the prior, where

$$H = \text{information} = \int \log(dP/dX) dP. \quad (11)$$

H is (minus) the logarithm of that fraction of prior mass that contains the bulk of the posterior mass, and it may well be of the order of thousands or more in practical problems where the likelihood is concentrated in some obscure corner of the prior domain.

To illuminate the width over X of the posterior, suppose the likelihood function has C approximately-Gaussian principal components, so that L is approximately a rank- C multivariate normal. For this model case, the likelihood can be written $L \propto \exp(-\frac{1}{2}r^2)$, where r is radius in C dimensions, with enclosed prior mass $X \propto r^C$. This posterior $dP \propto L dX = LX d \log X$ induces a standard deviation $\langle (\delta \log X)^2 \rangle^{1/2}$ of $\sqrt{C/2}$. So we expect the posterior mass to be fairly broadly distributed over a range something like $-H \pm \sqrt{C}$ in $\log X$. Generally, each useful principal component of the likelihood significantly restricts the range originally permitted by the prior (otherwise it's not useful), so H should usually exceed C , let alone $\sqrt{C}$, suggesting that locating and reaching the posterior domain is a more difficult task than navigating within it. This qualitative behaviour, where the posterior mass is mostly around $\log X \approx -\text{Huge} \pm \text{big}$ (Huge meaning H), is widely seen in practical applications.

To cover such a range, sampling ought to be linear in $\log X$ rather than in X , and we set

$$X_1 = t_1, \quad X_2 = t_1 t_2, \quad \dots, \quad X_i = t_1 t_2 \dots t_i, \quad \dots, \quad X_m = t_1 t_2 t_3 \dots t_m, \quad (12)$$

where each t_i lies between 0 and 1. It is these ratios t that control the subsequent calculations. If, for example, we could set $t = 0.99$ each time, then we should reach the bulk of the posterior after something like $100H$ steps, and cross it in a further $100\sqrt{C}$ steps. Any such sequence t leads to an estimate of Z , which we can make explicit by writing

$$Z(\mathbf{t}) \longleftarrow \sum_{i=1}^m w_i(\mathbf{t}) L_i. \quad (13)$$

5 Nested sampling idea

Although we cannot usually set precise values of t , it turns out that we can often set them statistically, and that is enough. All we need do, at step i , is take a random new point X_i from the prior, subject to $X_i < X_{i-1}$ (starting with $X_0 = 1$). Our knowledge of the new point $X_i = t_i X_{i-1}$ would be specified by $t_i \in \text{Uniform}(0, 1)$.

尚未找到的值并不重要。在此前提下，上限（10）仍然成立，数值积分的误差最多为 $\mathcal{O}(N^{-1})$ —，即上限与下限之差。

梯形规则 $w_i = \frac{1}{2}(X_{i-1} - X_{i+1})$ ，通过使用反射边界条件 $X_0 = 2 - X_1$ 和 $X_{m+1} = -X_m$ 来避免不合适的行为，在大多数情况下将其简化为 $\mathcal{O}(N^{-2})$ 。被积函数已经相当良好，因此我们预计不会比梯形规则（或类似规则）有显著改进，该规则由 “ $\longleftarrow$ ” 箭头表示。

4 证据

对于 Z 的积分，主要取决于后验质量的大部分所在的位置。通常，这部分仅占先验的一小部分 e^{-H} ，其中

$$H = \text{information} = \int \log(dP/dX) dP. \quad (11)$$

H 是包含后验质量大部分的先验质量分数（负）对数，在似然性集中在先验域某个隐晦角落的实际问题中，它可能达到数千或更多。

为了照亮后方的宽度超过 X ，假设似然函数有 C 近似高斯的主成分，使得 L 近似为秩- C 多元正态。对于这个模型情况，似然可以写成 $L \propto \exp(-\frac{1}{2}r^2)$ ，其中 r 是 C 维度的半径，包含的先验质量为 $X \propto r^C$ 。这个后验 $dP \propto L dX = LX d \log X$ 对数 X 产生一个标准差 $\langle (\delta \log X)^2 \rangle^{1/2}$ 为 $\sqrt{C/2}$ 。因此，我们预期后验质量会在某个类似 $-H \pm \sqrt{C}$ 的对数 X 范围内相当广泛地分布。通常，每个有用的似然主成分会显著限制先验最初允许的范围（否则它就无用），所以 H 通常会超过 C ，更不用说 $\sqrt{C}$ ，这表明定位和达到后验域比在其内部导航是一个更难的任务。这种定性行为，即后验质量主要围绕对数 $X \approx -\text{Huge} \pm \text{big}$ （Huge 的意思是 H ），在实践中被广泛观察到。

要覆盖如此范围，采样应该在 $\log X$ 上是线性的，而不是在 X 上，我们设定

$$X_1 = t_1, \quad X_2 = t_1 t_2, \quad \dots, \quad X_i = t_1 t_2 \dots t_i, \quad \dots, \quad X_m = t_1 t_2 t_3 \dots t_m, \quad (12)$$

其中每个 t_i 都介于 0 和 1 之间。正是这些比率 t 控制了后续计算。例如，如果我们每次都能设定 $t = 0.99$ ，那么我们大约需要 $100H$ 步就能达到后验分布的大部分，并在进一步的 $100\sqrt{C}$ 步中穿过它。任何这样的序列 t 都会导致对 Z 的估计，我们可以通过写成以下形式来使其明确

$$Z(\mathbf{t}) \longleftarrow \sum_{i=1}^m w_i(\mathbf{t}) L_i. \quad (13)$$

5 嵌套采样思想

虽然我们通常无法设定 t 的精确值，但结果发现我们通常可以统计性地设定它们，这就足够了。在步骤 i 中，我们只需要从先验分布中随机取一个新的点 X_i ，并受 $X_i < X_{i-1}$ 约束（从 $X_0 = 1$ 开始）。我们对新点 $X_i = t_i X_{i-1}$ 的了解将由 $t_i \in \text{Uniform}(0, 1)$ 指定。

In principle, such a point could be obtained by sampling X_i uniformly from within the corresponding restricted range $(0, X_{i-1})$, then interrogating the original likelihood-sorting to discover what its θ_i would have been. In practice, it would naturally be obtained directly as θ_i , by sampling within the equivalent constraint $L(\theta) > L_{i-1}$ (with $L_0 = 0$ to ensure complete initial coverage), in proportion to the prior density $\pi(\theta)$. This too finds a random point, distributed just the same. The second method is equivalent to the first, but bypasses the use of X . **So we don't need to do the sorting after all!** That's the key.

Successive points are illustrated in Figure 3, in which prior mass is represented by area. Thus, point 2 is found by sampling over the prior within the box defined by $L > L_1$, and so on. Such points will usually be found by some MCMC approximation, starting at some point θ^* known to obey the constraint (if available), or at worst starting at θ_{i-1} which lies on and defines the current likelihood boundary.

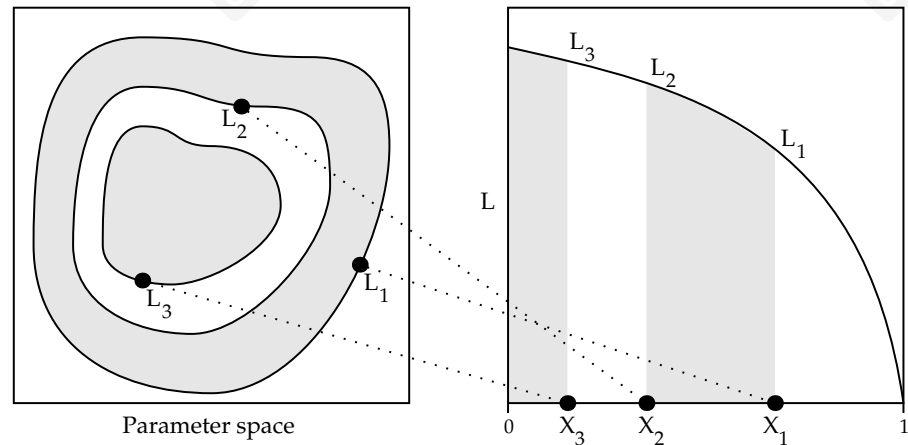


Figure 3: Nested likelihood contours are sorted to enclosed prior mass X .

It is not the purpose of this introductory paper to develop the technology of navigation within such a volume. We merely note that exploring a hard-edged likelihood-constrained domain should prove to be neither more nor less demanding than exploring a likelihood-weighted space. For example, consider a uniform prior weighted by a C -dimensional unit Gaussian likelihood $L(\theta) = \exp(-\frac{1}{2}|\theta|^2)$. Conventional Metropolis-Hastings exploration is simply accomplished with trial moves of arbitrary direction having step-length $|\delta\theta|$ around 1 for efficiency. Most points have $|\theta| \approx \sqrt{C}$, so the relaxation time is about C steps.

In nested sampling, the corresponding hard constraint is the ball $|\theta| < \sqrt{C}$ (or thereabouts). The typical point has $|\theta| \approx \sqrt{C} - 1/\sqrt{C}$, that being the median radius of the ball. Again, the efficient trial step-length is $|\delta\theta| \approx 1$, so the relaxation time per iterate is much the same as before. There are well-developed methods, such as Hamiltonian (or “hybrid”) Monte Carlo (Duane et al. (1987), Neal (1993)), slice sampling (Neal (2003)) and more, which learn about more general shapes of L in order to explore the likelihood-weighted space more efficiently. Similar methods ought to work for exploring likelihood-constrained domains, but have not yet been developed.

In terms of prior mass, successive intervals w scan the prior range from $X = 1$ down to

原则上，这样的点可以通过在相应的受限范围内均匀采样 X_i （即 $0, X_{i-1}$ ）来获得，然后通过查询原始的似然排序来发现它的 θ_i 会是什么样的。在实践中，它自然会直接作为 θ_i 获得，通过在等效约束 $L(\theta) > L_{i-1}$ （并使用 $L_0 = 0$ 确保完全的初始覆盖）内采样，与先验密度 π （即 θ ）成正比。这也找到了一个随机点，分布完全相同。第二种方法是第一种方法的等价物，但它绕过了使用 X 。所以我们根本不需要进行排序！这就是关键。

后续点在图3中进行了说明，其中先验质量由面积表示。因此，点2是通过在由6定义的框内对先验进行抽样找到的，等等。此类点通常通过某种MCMC近似找到，从某个已知服从约束的点 θ^* （如果可用）开始，或者在最坏情况下从位于并定义当前似然边界的点 θ_{i-1} 开始。

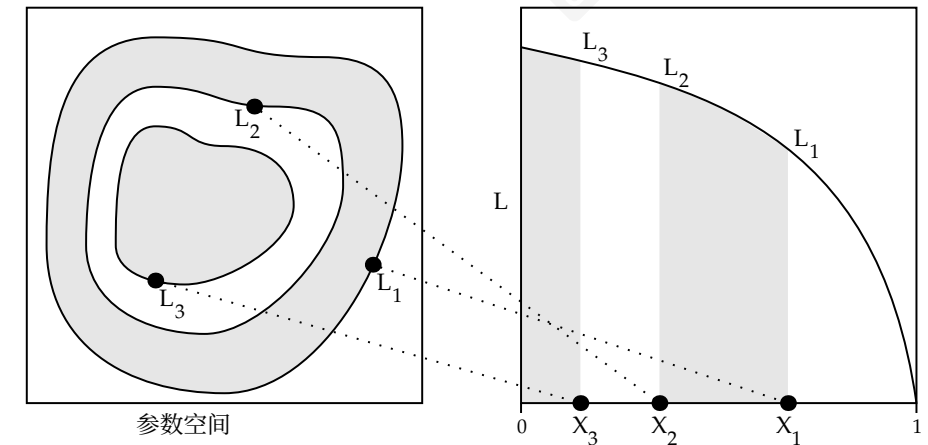


图3：嵌套似然轮廓按顺序包含先验质量 X 。

本文的引言目的并非开发在如此体积内导航的技术。我们仅仅指出，探索具有硬边界的似然约束域应该既不比探索似然加权空间更困难，也不更简单。例如，考虑一个由 C -维单位高斯似然 $L(\theta) = \exp(-\frac{1}{2}|\theta|^2)$ 加权的均匀先验。常规的Metropolis-Hastings探索通过具有步长 $|\delta\theta|$ （约为1以实现效率）的任意方向试验移动来简单地完成。大多数点具有 $|\theta| \approx \sqrt{C}$ ，因此弛豫时间约为 C 步。

在嵌套采样中，相应的硬约束是球 $|\theta| < \sqrt{C}$ （或附近）。典型点有 $|\theta| \approx \sqrt{C} - 1/\sqrt{C}$ ，即球的中间半径。同样，有效的试验步长是 $|\delta\theta| \approx 1$ ，因此每次迭代的松弛时间与之前大致相同。存在一些成熟的方法，例如哈密顿（或混合）蒙特卡洛（Duane 等人 (1987), Neal (1993)）、切片采样（Neal (2003)）等更多方法，它们学习更一般的 L 形状，以便更有效地探索加权似然空间。类似的方法应该适用于探索似然约束域，但尚未开发出来。

在先验质量方面，连续区间 w 扫描先验范围从 $X = 1$ 到

$X = 0$. In terms of coordinates θ , the intervals represent nested shells around contours of constant likelihood value, with points exactly on the same contour being ranked by their labels ℓ . More generally, instead of taking one point within the likelihood-constrained box, take N of them where N is any convenient number, and select the worst (lowest L , highest X), as the i 'th point. This recurrence is

$$X_0 = 1, \quad X_i = t_i X_{i-1}, \quad \Pr(t_i) = N t_i^{N-1} \text{ in } (0, 1), \tag{14}$$

t_i being the largest of N random numbers from **Uniform**(0,1). The mean and standard deviation of $\log t$ (which dominates the geometrical exploration) are

$$\mathbf{E}(\log t) = -1/N, \quad \mathbf{dev}(\log t) = 1/N. \tag{15}$$

The individual $\log t$ are independent, so after i steps, the prior mass is expected to shrink to $\log X_i \approx -(i \pm \sqrt{i})/N$. Thus we expect the procedure to take about $NH \pm \sqrt{NH}$ steps to shrink down to the bulk of the posterior, and a further $N\sqrt{C}$ or so steps to cross it. For a crude implementation, we can simply proclaim $\log X_i = -i/N$ as if we knew it, though it's more professional to acknowledge the uncertainties.

Actually, it is not necessary to find N points anew at each step, because $N-1$ of them are already available, being the survivors after deleting the worst. Only one new point is required per step, and this θ may be found by any method that draws from the prior subject to $L(\theta)$ being above its constraint L_{i-1} . One method is to replace the deleted point by a copy of a random survivor, evolved within the box by MCMC for some adequate number of trials. Surviving points could be used as stationary guides in such exploration. Another method might be generation of new points by genetic mixing of the survivors' coordinates. All that matters is that the step ends with N usably independent samples within the box.

6 Nested sampling procedure

At each step, the procedure has N points $\theta_1, \dots, \theta_N$, with corresponding likelihoods $L(\theta_1), \dots, L(\theta_N)$, augmented to L^+ as in (6) if ties of likelihood are anticipated. The lowest (minimum) such value is the likelihood L_i associated with step i . There are to be j iterative steps.

```
Start with  $N$  points  $\theta_1, \dots, \theta_N$  from prior;
  initialise  $Z = 0, X_0 = 1$ .
Repeat for  $i = 1, 2, \dots, j$ ;
  record the lowest of the current likelihood values as  $L_i$ ,
  set  $X_i = \exp(-i/N)$  (crude) or sample it to get uncertainty,
  set  $w_i = X_{i-1} - X_i$  (simple) or  $(X_{i-1} - X_{i+1})/2$  (trapezoidal),
  increment  $Z$  by  $L_i w_i$ ,
  then replace point of lowest likelihood by new one drawn
    from within  $L(\theta) > L_i$ , in proportion to the prior  $\pi(\theta)$ .
Increment  $Z$  by  $N^{-1}(L(\theta_1) + \dots + L(\theta_N)) X_j$ .
```

The last step fills in the missing band $0 < X < X_j$ of the desired integral $\int_0^1 L dX$ with weight $w = N^{-1} X_j$ for each surviving point, after the iterative steps have compressed the domain

$X = 0$ 。在坐标 θ 方面，这些区间表示围绕恒定似然值等值线的嵌套壳层，位于同一等值线上的点按其标签 ℓ 排序。更一般地，与其在似然约束的框内取一个点，不如取 N 个点，其中 N 是任何方便的数字，并选择最差（最低 L ，最高 X ）的 i 个点。这种递归关系是

$$X_0 = 1, \quad X_i = t_i X_{i-1}, \quad \Pr(t_i) = N t_i^{N-1} \text{ in } (0, 1), \tag{14}$$

t_i 是来自均匀分布(0,1)的 N 个随机数中最大的。对数 t (主导几何探索)的均值和标准差是

$$\mathbf{E}(\log t) = -1/N, \quad \mathbf{dev}(\log t) = 1/N. \tag{15}$$

单个对数 t 是独立的，因此经过 i 步后，先验质量预计会缩小到对数 $X_i \approx -(i \pm \sqrt{i})/N$ 。因此我们预计该过程需要大约 $NH \pm \sqrt{NH}$ 步才能缩小到后验的主体，并且还需要大约 $N\sqrt{C}$ 步才能跨越它。对于粗糙的实现，我们可以简单地宣告对数 $X_i = -i/N$ ，就好像我们已知它一样，尽管承认不确定性更专业。

实际上，没有必要在每一步都重新找到 N 个点，因为其中 $N-1$ 个已经可用，它们是删除最差点后的幸存者。每一步只需要一个新点，并且这 θ 可以通过任何从先验中抽取且满足 $L(\theta)$ 高于其约束 L_{i-1} 的方法找到。一种方法是用随机幸存者的副本替换已删除的点，通过 MCMC 在盒子内进行足够数量的试验进行进化。幸存点可以用作此类探索中的固定引导。另一种方法可能是通过幸存者坐标的遗传混合生成新点。重要的是，该步骤结束时，盒子内必须有可用地独立的样本。

6 嵌套抽样程序

每一步，该程序有 N 个点 $\theta_1, \dots, \theta_N$ ，带有相应的似然值 $L(\theta_1), \dots, L(\theta_N)$ ，如果预期似然值会相等，则增加到 L^+ （如公式6所示）。这样的最低（最小）值是步骤 i 相关的似然值 L_i 。需要进行 j 次迭代步骤。

```
Start with  $N$  points  $\theta_1, \dots, \theta_N$  from prior;
  initialise  $Z = 0, X_0 = 1$ .
Repeat for  $i = 1, 2, \dots, j$ ;
  record the lowest of the current likelihood values as  $L_i$ ,
  set  $X_i = \exp(-i/N)$  (crude) or sample it to get uncertainty,
  set  $w_i = X_{i-1} - X_i$  (simple) or  $(X_{i-1} - X_{i+1})/2$  (trapezoidal),
  increment  $Z$  by  $L_i w_i$ ,
  then replace point of lowest likelihood by new one drawn
    from within  $L(\theta) > L_i$ , in proportion to the prior  $\pi(\theta)$ .
将  $Z$  增加  $N^{-1}(L(\theta_1) + \dots + L(\theta_N)) X_j$ 。
```

最后一步 将填充缺失的波段 $0 < X < X_j$ ，以获得所需的积分 $\int_0^1 L dX$ ，并为每个幸存的点赋予权重 $w = N^{-1} X_j$ ，在迭代步骤压缩了域之后

to X_j . So the final number of terms in the evidence summation (8) is $m = j + N$. However, there should already have been sufficient steps j to accumulate most of the integral, so this final increment ought to be an unimportant refinement. It is also possible to accumulate the information H along with Z .

Figure 4 illustrates the method, running with $N = 3$ points. Initially, three samples are taken from the unconstrained prior, whose mass is represented by the complete square area on the left. These points could equivalently have been taken randomly from X in $(0,1)$, as shown on the lower line on the right. They have labels 1, 3, 4, as yet unknown. In step 1, the worst (lowest L , highest X) of these points is identified as number 1, with likelihood L_1 . It is then replaced by a new point, drawn from inside the contour $L(\theta) > L_1$. Equivalently, it could have been taken randomly from X in $(0, X_1)$. Including the two survivors, there are still three samples, now all uniform in the reduced range $(0, X_1)$. With the particular random numbers used for Figure 4, the new point in step 1 happened to lie outside the two survivors, so became number 2, but in step 2 the new point happened to be the innermost, and was eventually identified as number 5. After the $j = 5$ allotted steps, the five discarded points 1,2,3,4,5 are augmented with the final three survivors 6,7,8 to give the $m = 8$ points $(X_1, \dots, X_8)$ shown on the top line. It is over these points that the sum $\sum_{i=1}^8 L_i w_i$ is evaluated to estimate Z .

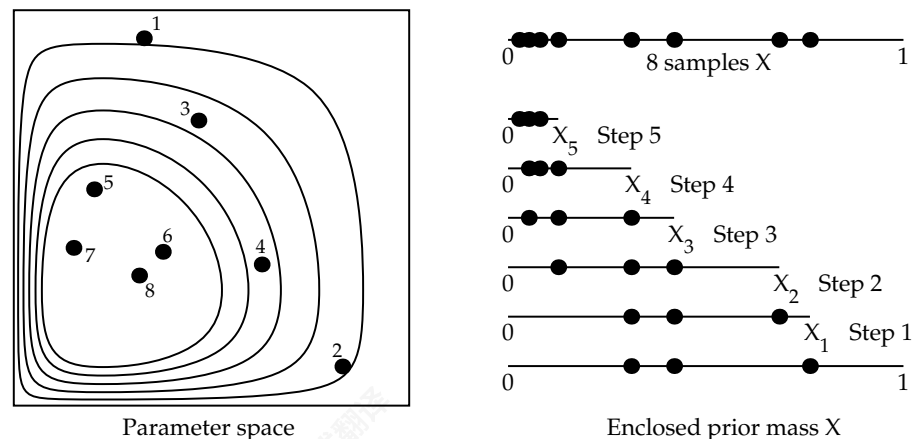


Figure 4: Nested sampling for five steps with a collection of three points. Likelihood contours shrink by factors $\exp(-1/3)$ in area and are roughly followed by successive sample points.

With $N = 3$ samples, shrinkage is expected to be roughly geometrical, by $\Delta \log X \sim -1/3$ per step. The diagram on the left of Figure 4 shows likelihood contours drawn at levels corresponding to enclosed areas diminishing by this factor — *i.e.* the i 'th contour encloses prior mass $e^{-i/3}$. Indeed, the first point lies close to the first contour, the second point is not too far outside the second contour, and so on until the fifth point chances to fall inside the fifth contour. If we could arrange exact matching, we would know the X 's and have a definitive answer for Z , depending only on the scheme of numerical integration. Since we can't arrange this, we proceed towards deriving a probabilistic estimate instead. Although it is unusual to deal with an integral in which the ordinate L is known and the abscissa X is uncertain, rather than the other way around, the problem remains soluble.

到 X_{j_0} 因此, 证据求和 (8) 中的项数 $\hat{n}$ 最终为 $m = j + N$ 。然而, 应该已经进行了足够的步骤 $\hat{n}$ 来累积大部分积分, 所以这个 $\hat{n}$ 最终增量应该是一个无关紧要的结果 $\hat{n}_0$ 。也可以沿着 H 累积信息 Z_0 。

图4说明了该方法, 使用 $N = 3$ 个点运行。最初, 从无约束先验中抽取三个样本, 其质量由左侧的完整正方形面积表示。这些点等效地可以从 X 在 $(0,1)$ 中随机抽取, 如图右侧下方的线所示。它们有标签1、3、4, 目前尚不清楚。在步骤1中, 这些点中最差 (最低 L , 最高 X) 的点被识别为编号1, 其似然度为 L_1 。然后它被一个新的点替换, 该点从轮廓 $L(\theta) > L_1$ 内部抽取。等效地, 它也可以从 X 在 $(0, X_1)$ 中随机抽取。包括两个幸存者, 仍然有三个样本, 现在都在缩小的范围内 $(0, X_1)$ 均匀分布。使用图4的特定随机数, 步骤1中的新点碰巧位于两个幸存者之外, 因此成为编号2, 但在步骤2中, 新点碰巧是最内部的, 最终被识别为编号5。经过 $j = 5$ 分配的步骤后, 被丢弃的点1、2、3、4、5被最后的三个幸存者6、7、8补充, 以给出顶线上显示的 $(X_1, \dots, X_8)$ 点。正是这些点用于评估总和 $\sum_{i=1}^8 L_i w_i$, 以估计 Z 。

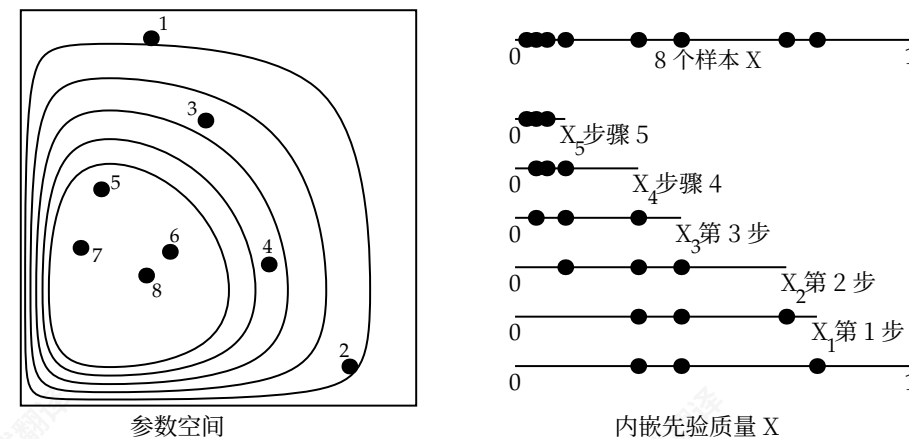


图4: 对 $\hat{n}$ 步进行嵌套采样, 包含三个点集。似然等高线在面积上缩小了 $\exp(-1/3)$ 倍, 并大致被后续采样点跟踪。

使用 $N = 3$ 个样本, 收缩预期将大致呈几何级数, 每步减少 $\Delta \log X \sim -1/3$ 。图4左侧的图表显示了对应于被包含面积按此因子 — *i.e.* 减少的 i 'th 等高线, 该等高线包含先验质量 $e^{-i/3}$ 。实际上, 第 $\hat{n}$ 个点靠近第 $\hat{n}$ 个等高线, 第二个点并未远离第二个等高线, 以此类推, 直到第 $\hat{n}$ 个点偶然落入第 $\hat{n}$ 个等高线内。如果我们能够实现精确匹配, 我们将知道 X 并对 Z 得出明确答案, 这仅取决于数值积分方案。由于我们无法实现这一点, 我们转而推导概率估计。虽然处理一个纵坐标 L 已知而横坐标 X 不确定而非反之的积分通常不常见, 但问题仍然是可解的。

7 Nested sampling termination

Termination of the main loop could simply be after a pre-set number of steps, or could be when even the largest current likelihood, taken over the full current box, would not increase the current evidence by more than some small fraction f ;

$$\max(L_1, \dots, L_N) X_j < f Z_j \implies \text{termination.} \quad (16)$$

Plausibly, the accumulation of Z is then tailing off, so the sum is nearly complete.

The usual behaviour of the areas $L_i w_i$ is that they start by rising, with the likelihood L_i increasing faster than the widths w_i decrease. The more important regions are being found. At some point, L flattens off sufficiently that decreasing width dominates increasing likelihood, so that the areas pass across a maximum and start to fall away. Most of the total area is usually found in the region of this maximum, which occurs in the region of $X \approx e^{-H}$. There are counter-examples, but that behaviour is often expected. Remembering $X_i \approx e^{-i/N}$, this suggests an alternative termination condition

$$\text{“continue iterating until the count } i \text{ significantly exceeds } NH\text{”,} \quad (17)$$

which still expresses the general aim that a nested-sampling calculation should be continued until most of Z has been found. (Of course, H is here the current evaluate from the previous i iterates.) This is the criterion used in the Appendix program.

Unfortunately, we can offer no rigorous criterion based on sampling to ensure the validity of any such termination condition. It is perfectly possible for the accumulation of Z to flatten off, apparently approaching a final value, whilst yet further inward there lurks a small domain in which the likelihood is sufficiently large to dominate the eventual results. Termination remains a matter of user judgment about the problem at hand, albeit with the aim of effectively completing the accumulation of Z . If in doubt, continue upward and inward.

However, if an analytical upper bound $L \leq L_{\max}$ can be found, such as when a Gaussian likelihood factor cannot exceed $1/\sqrt{2\pi}\sigma$, it can be used in (16) to give a firmer termination criterion

$$L_{\max} X_j < f Z_j \implies \text{termination.} \quad (18)$$

In this case, all but a fraction f of Z has been found.

8 Numerical uncertainty

It is possible to run nested sampling crudely, by assigning each $\log t$ its mean value of $-1/N$, and ignoring its uncertainty. With X_i thereby being set to $e^{-i/N}$, this captures the basic idea by giving a quick picture of the likelihood function $L(X)$. An early example of a similar approach is [McDonald and Singer \(1967\)](#), and it is encoded in the simple Appendix program. However, a fuller and better treatment is also possible.

For a given choice of coefficients $\mathbf{t}$, the estimate of Z would be $\sum_i L_i w_i(\mathbf{t})$ from (13). One such choice of $\mathbf{t}$ will be correct, corresponding to the selected points θ_i , but we do not know which, and it could have been any. In particular, the correct choice of $\log t_i$ is most unlikely to be $-1/N$ for all i . Instead, the “sequence probability” $\Pr(\mathbf{t}) d\mathbf{t} = \prod_i N t_i^{N-1} dt_i$ from (14) induces a distribution for our estimates of Z ;

$$\Pr(Z) \longleftarrow \int \delta\left(Z - \sum_{i=1}^m L_i w_i(\mathbf{t})\right) \Pr(\mathbf{t}) d\mathbf{t}. \quad (19)$$

7 嵌套抽样终止

主循环的终止可以简单地设置为预设的步数之后，或者可以在当前整个盒子中即使最大的当前似然也不会使当前证据增加超过某个小分数 f 时发生；

$$\max(L_1, \dots, L_N) X_j < f Z_j \implies \text{termination.} \quad (16)$$

可能地， Z 的累积随后开始尾部 off，所以总和几乎已经完成。

区域 $L_i w_i$ 的通常行为是它们先上升，似然 L_i 增加得比宽度 w_i 减少得更快。更重要的区域正在被发现。在某个时刻， L 变得 off suffi 足够，以至于减少宽度主导增加似然，所以区域穿过一个最大值并开始减少。通常大部分总面积是在这个最大值的区域中，它发生在区域 $X \approx e^{-H}$ 中。有反例，但这种行为通常是预期的。记住 $X_i \approx e^{-i/N}$ ，这表明一个替代的终止条件

$$\text{“continue iterating until the count } i \text{ significantly exceeds } NH\text{”,} \quad (17)$$

这仍然表达了嵌套采样计算应持续进行，直到找到大部分 Z 的总体目标。（当然，这里的 H 是来自先前 i 迭代中的当前评估值。）这是附录程序中使用的标准。

不幸的是，我们无法基于采样提供任何严格的准则来确保任何此类终止条件的有效性。 Z 的累积完全可能 $\nrightarrow$ 衰减，明显接近一个 $\nrightarrow$ 终值，而此时在更深处仍潜伏着一个可能性足够大的小区域，足以主导最终结果。终止仍然是一个关于所处理问题的用户判断问题，尽管目标是有效地完成 Z 的累积。如有疑问，请继续向上并向内探索。

然而，如果能找到一个解析上限 $L \leq L_{\max}$ ，例如高斯似然因子不能超过 $1/\sqrt{2\pi}\sigma$ 时，它可以用于 (16) 以给出一个更明确的终止标准。

$$L_{\max} X_j < f Z_j \implies \text{termination.} \quad (18)$$

在这种情况下，除了 f 的 Z 的一部分之外，其余部分均已找到。

8 数值不确定性

可以通过为每个对数 t 分配其 $-1/N$ 的平均值并忽略其不确定性来粗略地运行嵌套采样。当 X_i 被设置为 $e^{-i/N}$ 时，这通过给出似然函数 $L(X)$ 的快速图像来捕捉基本思想。一个类似的早期例子是麦当劳和辛格（1967年），并且它被编码在简单的附录程序中。然而，也可以进行更全面和更好的处理。

对于给定的系数 $\mathbf{t}$ 的选择， Z 的估计将是 $\sum_i L_i w_i(\mathbf{t})$ 根据 (13)。这样的 $\mathbf{t}$ 选择之一将是正确的，对应于选定的点 θ_i ，但我们不知道是哪一个，它可能是任何值。特别是，正确的 $\log t_i$ 选择极不可能对所有 i 都等于 $-1/N$ 。相反，来自 (14) 的“序列概率” $\Pr(\mathbf{t}) d\mathbf{t} = \prod_i N t_i^{N-1} dt_i$ 为我们的 Z 估计诱导了一个分布；

$$\Pr(Z) \longleftarrow \int \delta\left(Z - \sum_{i=1}^m L_i w_i(\mathbf{t})\right) \Pr(\mathbf{t}) d\mathbf{t}. \quad (19)$$

Analytic expressions for moments $\langle Z \rangle, \langle Z^2 \rangle, \dots$ are available through means, correlations, and higher moments of $t_1, t_1 t_2, t_1 t_2 t_3, \dots$. However, the dominant uncertainty in Z is usually due to the Poisson variability $NH \pm \sqrt{NH}$ in the number of steps to reach the posterior, implying a geometrical uncertainty factor $\exp(\pm \sqrt{H/N})$ which could be many powers of e . Hence it is $\log Z$ rather than Z itself that should usually have a fairly symmetrical roughly-normal uncertainty. The distribution of Z may be awkwardly skewed, with its moments dominated by occasional upward outliers in $\log Z$. A reader who doubts this may consider the hypothetical normally-distributed $\log Z = 100 \pm 10$, whose direct mean and standard deviation $Z = e^{150} \pm e^{200}$ are unhelpful, especially since it is known that Z has to be positive. So, rather than proceeding with moment expansions, it seems better to use Monte Carlo, taking a set $\{\mathbf{t}\}$ of several dozen samples from the sequence probability $\Pr(\mathbf{t})$ to simulate the X 's and thence obtain the distribution of Z from samples $\{Z\}_{\mathbf{t}}$, from which the statistics of $\log Z$ can be read off with adequate confidence:

$$\log Z \longleftarrow \text{estimate} \pm \text{uncertainty, from } \{\log Z\}_{\mathbf{t}}. \quad (20)$$

The uncertainty accompanying such estimates will usually diminish as the inverse square root of N , the amount of computation that one is prepared to invest in the original exploration. The same approach can be applied to any average over $\mathbf{t}$:

$$\int \dots \Pr(\mathbf{t}) d\mathbf{t} = \langle \dots \rangle_{\mathbf{t}}. \quad (21)$$

Alongside the uncertainty of about $\pm \sqrt{H/N}$ in $\log Z$, there is also the systematic numerical bias imposed by the integration rule. Thus, the lower- or upper-bound numerical weights (9) or (10) will usually give $\mathcal{O}(1/N)$ bias, which the trapezoidal improvement reduces to $\mathcal{O}(1/N^2)$. Usually, these biases are overwhelmed by the uncertainty.

Technically, $\Pr(Z)$ is most easily evaluated by Monte Carlo sampling over $\mathbf{t}$. That is highly unlikely to be a significant source of error if several dozen samples are used (which is allowable because no likelihood evaluations are involved). However, if this is thought to be of concern, it is worth noting that Monte Carlo could be evaded by evaluating the integral differently. Write (13) (with lower-bound weights (9) for simplicity) as

$$Z(\mathbf{t}) \longleftarrow \sum_{i=1}^m \lambda_i X_i(\mathbf{t}), \quad \lambda_i = L_i - L_{i-1} \quad (22)$$

with $L_0 = 0$, and expand X_i using (12):

$$Z(\mathbf{t}) \longleftarrow t_1(\lambda_1 + t_2(\lambda_2 + \dots + t_{m-1}(\lambda_{m-1} + t_m(\lambda_m)) \dots)). \quad (23)$$

Working outwards, use the recurrence relation

$$Z_m = \lambda_m, \quad Z_{i-1} = \lambda_{i-1} + t_i Z_i \quad (24)$$

to reach the required $Z = Z_1$. Each step of this takes a computed distribution $\Pr(Z_i)$ and integrates it with $\Pr(t)$ from (14) (a convolution over $\log X$) before shifting it by λ_{i-1} to reach $\Pr(Z_{i-1})$, ending with the required $\Pr(Z)$, from which statistics of Z or $\log Z$ can be read off. In this way, the m -dimensional integral (19) can be reduced to m feasible one-dimensional operations without any methodological qualms.

关于矩 $\langle Z \rangle, \langle Z^2 \rangle, \dots$ 的解析表达式可以通过 $t_1, t_1 t_2, t_1 t_2 t_3, \dots$ 的均值、相关性和高阶矩获得。然而, Z 中的主要不确定性通常是由于到达后验所需的步数中的泊松变异性 $NH \pm \sqrt{NH}$, 这意味着一个几何不确定性因子 $\exp(\pm \sqrt{H/N})$, 它可能是 e 的许多幂。因此, 通常是 $\log Z$ 而不是 Z 本身应该具有相当对称的大致正态不确定性。 Z 的分布可能 awkwardly 偏斜, 其矩由 $\log Z$ 中偶尔的上行离群值主导。一个怀疑这一点的人可以考虑假设的正态分布 $\log Z = 100 \pm 10$, 其直接均值和标准差 $Z = e^{150} \pm e^{200}$ 没有帮助, 尤其是由于已知 Z 必须为正。因此, 与其进行矩展开, 不如使用蒙特卡洛, 从序列概率 $\Pr(\mathbf{t})$ 中取几十个样本 $\{\mathbf{t}\}$ 来模拟 X 的, 然后从样本 $\{Z\}_{\mathbf{t}}$ 获得分布 Z , 从而以足够的 confidence 读取 $\log Z$ 的统计数据:

$$\log Z \longleftarrow \text{estimate} \pm \text{uncertainty, from } \{\log Z\}_{\mathbf{t}}. \quad (20)$$

此类估计的不确定性通常会随着愿意投入原始探索的计算量的倒数平方根 N 而减小。同样的方法可以应用于 $\mathbf{t}$ 的任何平均值:

$$\int \dots \Pr(\mathbf{t}) d\mathbf{t} = \langle \dots \rangle_{\mathbf{t}}. \quad (21)$$

除了大约 $\mathbf{t} \pm \sqrt{H/N}$ 的 $\log Z$ 不确定性之外, 还存在由积分规则引入的系统数值偏差。因此, 下限或上限数值权重 (9) 或 (10) 通常会给出 $\mathcal{O}(1/N)$ 偏差, 而梯形改进将此偏差减少到 $\mathcal{O}(1/N^2)$ 。通常, 这些偏差会被不确定性所淹没。

技术上, $\Pr(Z)$ 最容易通过在 $\mathbf{t}$ 上进行蒙特卡洛采样来评估。如果使用几十个样本 (这是允许的, 因为没有涉及似然评估), 那么这不太可能成为显著的误差来源。然而, 如果认为这是一个问题, 值得注意的是, 可以通过以不同的方式评估积分来避免蒙特卡洛。将 (13) (为简单起见, 使用下限权重 (9)) 写为

$$Z(\mathbf{t}) \longleftarrow \sum_{i=1}^m \lambda_i X_i(\mathbf{t}), \quad \lambda_i = L_i - L_{i-1} \quad (22)$$

借助 $L_0 = 0$, 并扩展 X_i 使用 (12):

$$Z(\mathbf{t}) \longleftarrow t_1(\lambda_1 + t_2(\lambda_2 + \dots + t_{m-1}(\lambda_{m-1} + t_m(\lambda_m)) \dots)). \quad (23)$$

向外推导, 使用递推关系

$$Z_m = \lambda_m, \quad Z_{i-1} = \lambda_{i-1} + t_i Z_i \quad (24)$$

要达到所需的 $Z = Z_1$ 。每一步都采用计算分布 $\Pr(Z_i)$, 并将其与来自(14)的 $\Pr(t)$ (对 $\log X$ 的卷积) 进行整合, 然后通过 λ_{i-1} 进行平移以达到 $\Pr(Z_{i-1})$, 最终得到所需的 $\Pr(Z)$, 从中可以读取 Z 或 $\log Z$ 的统计数据 off。通过这种方式, m -维积分(19)可以被简化为 m 个可行的单维操作, 没有任何方法论上的疑虑。

9 Simple example

In C dimensions, consider the simple Gaussian problem

$$L(\theta) = \exp\left(-\frac{r^2}{2\sigma^2}\right), \quad r^2 = \sum_{i=1}^C \theta_i^2, \quad (25)$$

where θ has flat prior within the unit sphere

$$\text{prior } \pi(\theta) = \frac{(C/2)!}{\pi^{C/2}} \quad (\pi = 3.1415\dots) \quad \text{in } r < 1. \quad (26)$$

For convenience, take $\sigma \ll C^{-1/2}$, so that almost all the likelihood is well within the prior domain. The evidence (discarding the tails outside the domain) evaluates to

$$Z = (C/2)! (2\sigma^2)^{C/2}. \quad (27)$$

We observe that L is a decreasing function of radius r , so that sorting (were we able to perform it) would organise θ into an outward radial sequence of nested shells, having enclosed prior mass

$$X = r^C. \quad (28)$$

Hence the likelihood function is

$$L(X) = \exp(-X^{2/C}/2\sigma^2), \quad (29)$$

as plotted in Figure 5 for $C = 10$, $\sigma = 0.01$.

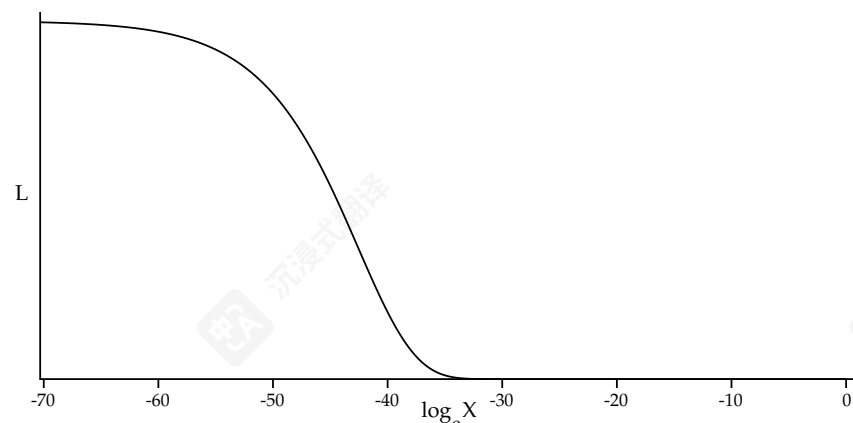


Figure 5: Likelihood function of simple example (on logarithmic abscissa).

The bulk of this mass is around radius $r \sim \sigma\sqrt{C}$, and the information required to reach down into this small patch is

$$H \approx -C \log(\sigma\sqrt{C}). \quad (30)$$

Meanwhile, the true numerical values are $\log Z = -37.81$, $H = 32.80$. (Although Figure 5 is a little misleading visually because of the logarithmic scale, the bulk of the posterior is indeed around $\log X \approx -H \approx -33$.)

9 简单示例

在 C 维度中, 考虑简单的高斯问题

$$L(\theta) = \exp\left(-\frac{r^2}{2\sigma^2}\right), \quad r^2 = \sum_{i=1}^C \theta_i^2, \quad (25)$$

其中 θ 在单位球域内有先验分布

$$\text{prior } \pi(\theta) = \frac{(C/2)!}{\pi^{C/2}} \quad (\pi = 3.1415\dots) \quad \text{in } r < 1. \quad (26)$$

为方便起见, 取 $\sigma \ll C^{-1/2}$, 这样几乎所有的似然度都很好地落在先验域内。证据 (舍弃域外的尾部) 评估为

$$Z = (C/2)! (2\sigma^2)^{C/2}. \quad (27)$$

我们观察到 L 是半径 r 的减函数, 因此 (如果我们能够执行排序) 会组织 θ 成一个向外径向排列的嵌套壳序列, 包含先验质量

$$X = r^C. \quad (28)$$

因此似然函数为

$$L(X) = \exp(-X^{2/C}/2\sigma^2), \quad (29)$$

如图5所示, 对于 $C = 10$ 和 $\sigma = 0.01$ 。

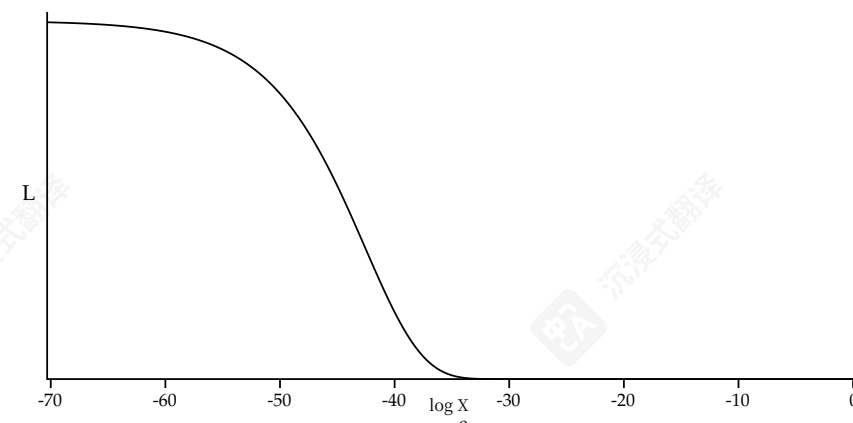


图5: 简单示例的似然函数 (纵轴为对数)。

这部分质量主要集中在大约半径 $r \sim \sigma\sqrt{C}$ 处, 而要深入到这个小区域所需的信息是

$$H \approx -C \log(\sigma\sqrt{C}). \quad (30)$$

与此同时, 真实的数值是 $\log Z = -37.81$, $H = 32.80$ 。 (尽管由于对数刻度, 图5在视觉上有些误导, 但后验分布的大部分确实在 $\log X \approx -H \approx -33$ 附近)

It is our task to reproduce this likelihood function, and thence the evidence as its integral, by nested sampling. For simplicity, consider using just one sample $N = 1$.

The initial step is to take a random point θ_1 within the unit sphere, and evaluate its likelihood $L_1 = L(\theta_1)$. The next step is to take a random point θ_2 of greater likelihood, $L(\theta_2) > L_1$. Using our analytical insight into this particular problem, we know that we can obtain such a point by sampling within the sphere $r < r_1$. We have special knowledge, and can use it. However, a nested sampling program would generally have to find θ_2 by some other method. It might use MCMC over the prior, starting at θ_1 and accepting only those points with greater likelihood, until correlation with the original point was deemed to be lost. Or it might use something else. We do not discuss such choices here; we simply assume that sampling from within a likelihood constraint is possible.

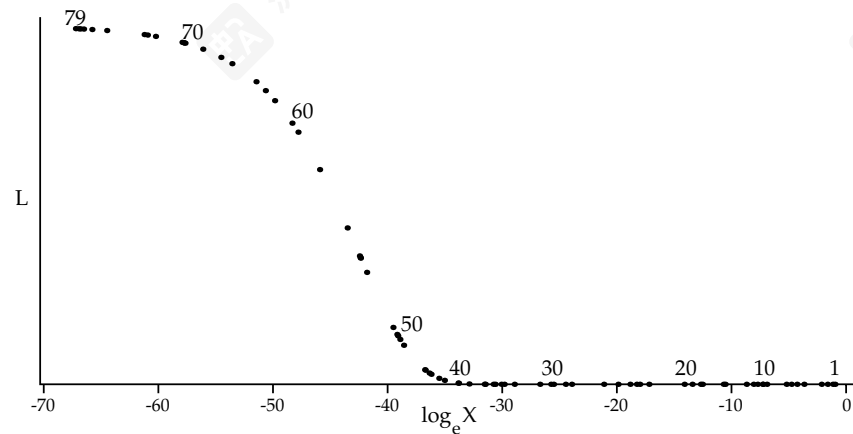


Figure 6: A random sequence of the first 79 nested sampling points, shown at their true X values. With $N = 1$, the i 'th point lies at $\log X = -i$ on average, but particular sequences drift randomly away.

The next step is to take a third random point θ_3 of yet greater likelihood, $L_3 = L(\theta_3) > L_2$, and so on indefinitely. If we could use our analytical insight (28), we would evaluate the enclosed prior masses X and hence determine the sequence $(X_1, L_1), (X_2, L_2), (X_3, L_3), \dots$ properly, as shown in Figure 6 for one particular run. By construction, these points lie exactly on the true likelihood curve. Although the points are quite coarsely spaced so that $\mathcal{O}(1)$ errors are expected, numerical integration (by trapezoid, say) gives a respectable estimate $\log Z \approx -37.60$. This example used just one point ($N = 1$), but for more accuracy we might evolve a collection of $N > 1$ points to obtain a sequence N times more closely spaced and N^2 more accurate. But, without the analytical insight available in this simple example, we would not have exact locations X , so we could not use this scheme.

However, we do have crude central estimates $\log X_i = -i/N$ for the locations, and could use them as in Figure 7. The general profile of the likelihood function is preserved, in that likelihood values L and their ordering are the same. However, the $\log X$ abscissa has been distorted to keep the points uniformly spaced in keeping with the central estimates. By the time the bulk of the posterior has been reached, the accumulated Poisson errors have shifted the peak away from $i \approx H$ by $\delta i \sim \pm\sqrt{i}$, so that $\log X$ has been offset, leading to a corresponding (and dominating) uncertainty of $\pm\sqrt{H}$ in $\log Z$ (or $\pm\sqrt{H/N}$ for general N). Numerically, this particular result was $\log Z = -43.6 \pm 5.9$, being these crude estimates of the mean and standard

我们的任务是使用嵌套采样来重现这个似然函数，进而通过积分得到其证据。为了简化，考虑仅使用一个样本 $N = 1$ 。

第一步是取单位球内的一个随机点 θ_1 ，并评估其似然 $L_1 = L(\theta_1)$ 。下一步是取一个似然更大的随机点 θ_2 ， $L(\theta_2) > L_1$ 。基于我们对这个特定问题的分析洞察，我们知道可以通过在球体 $r < r_1$ 内采样来获得这样的点。我们拥有特殊知识，并且可以使用它。然而，嵌套采样程序通常必须通过其他方法 θ_1 和 θ_2 。它可能使用先验分布上的 MCMC，从 θ_1 开始，只接受似然更大的点，直到认为与原始点的相关性被认为丢失为止。或者它可能使用其他方法。我们不在这里讨论这些选择；我们只是假设在似然约束内进行采样是可能的。

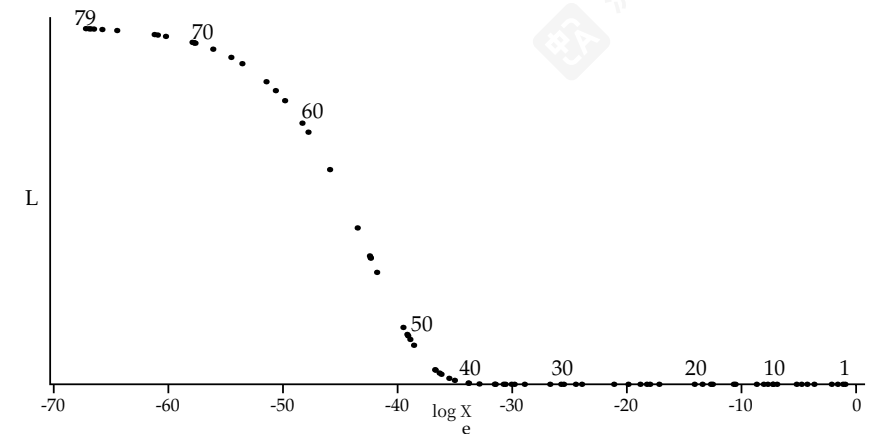


图6：展示 i 个嵌套采样点的随机序列，这些点位于其真实 X 值处。使用 $N = 1$ 时，第 i 个点平均位于 $\log X = -i$ 处，但特定序列会随机偏离。

下一步是取一个概率更大的第三个随机点 θ_3 ， $L_3 = L(\theta_3) > L_2$ ，等等无限进行下去。如果我们能运用分析洞察力 (28)，我们会评估所包含的先验质量 X ，从而正确确定序列 $(X_1, L_1), (X_2, L_2), (X_3, L_3), \dots$ ，如图6中的一次特定运行所示。按构造，这些点精确位于真实似然曲线上。尽管这些点间隔相当粗，预计存在 $\mathcal{O}(1)$ 误差，但数值积分（例如梯形法）可给出可观的估计 $\log Z \approx -37.60$ 。此例仅使用了一个点 ($N = 1$)，但为了更高精度，我们可能演化一个由 $N > 1$ 个点组成的集合，以获得间隔更紧密、更精确的序列 N 。但是，没有此简单示例中可用的分析洞察力，我们无法获得精确位置 X ，因此无法使用此方案。

然而，我们确实有关于位置的粗略中心估计 $\log X_i = -i/N$ ，可以像图7那样使用它们。似然函数的总体轮廓得以保留，因为似然值 L 及其排序保持不变。然而，为了使点在中心估计的范围内均匀分布， $\log X$ 横坐标已被扭曲。当后验分布的大部分已经达到时，累积的泊松误差已经将峰值从 $i \approx H$ 偏移了 $\delta i \sim \pm\sqrt{i}$ ，因此 $\log X$ 被抵消，导致 $\log Z$ (or $\pm\sqrt{H/N}$ 对一般 N) 出现相应的（且占主导地位的）不确定性 $\pm\sqrt{H}$ 。数值上，这个特定结果为 $\log Z = -43.6 \pm 5.9$ ，因为这些都是均值和标准偏差的粗略估计。

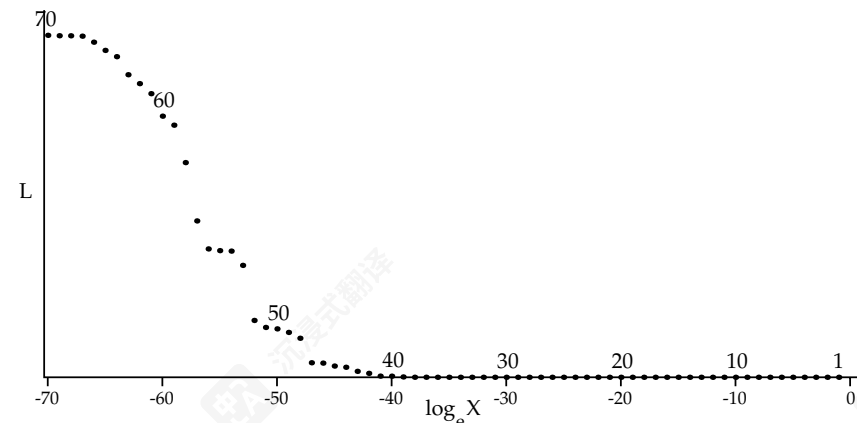


Figure 7: The same sequence of nested sampling points as in Fig. 6, shown at the crude central estimate of their X values (*i.e.* uniformly in $\log X$).

deviation.

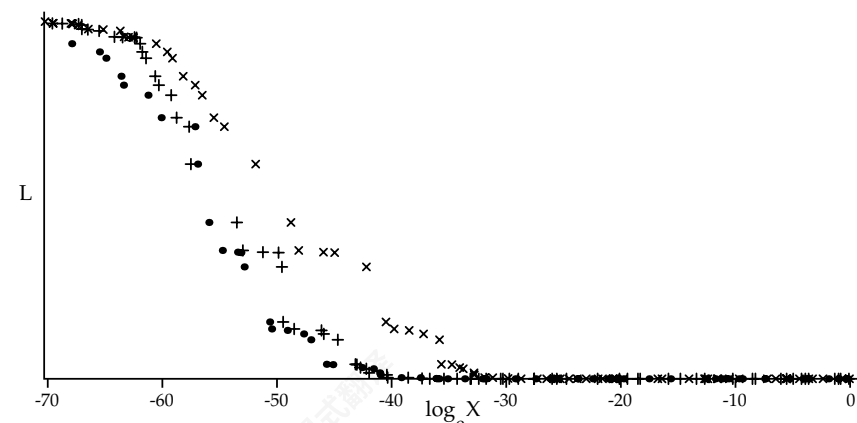


Figure 8: The same sequence of nested sampling points as in Fig. 6, showing three random assignments ($\bullet$, $+$, $\times$) for X values taken from $\text{Pr}(X)$.

A more professional but quite similar estimate $\log Z = -42.7 \pm 5.5$ (again mean and standard deviation) is obtained by sampling (*i.e.* repeatedly guessing) the mass ratios $\mathbf{t}$ from their pdf (14), and accumulating the values of Z that are thereby produced. Three of these guesses are shown in Figure 8. As expected, the curves are mostly the same shape and differ principally by random offset in $\log X$. Averaged over many assignments of $\mathbf{t}$, this better estimate misses the true value $\log Z = -37.81$ by an unremarkable 0.9 standard deviations.

It is, of course, necessary to take enough iterates (at least NH) to reach and then cross the bulk of the posterior mass. Likelihood values can guide the termination decision by indicating when the accumulation of Z appears to be tailing off, perhaps as in (17), but there is seldom a definitive criterion.

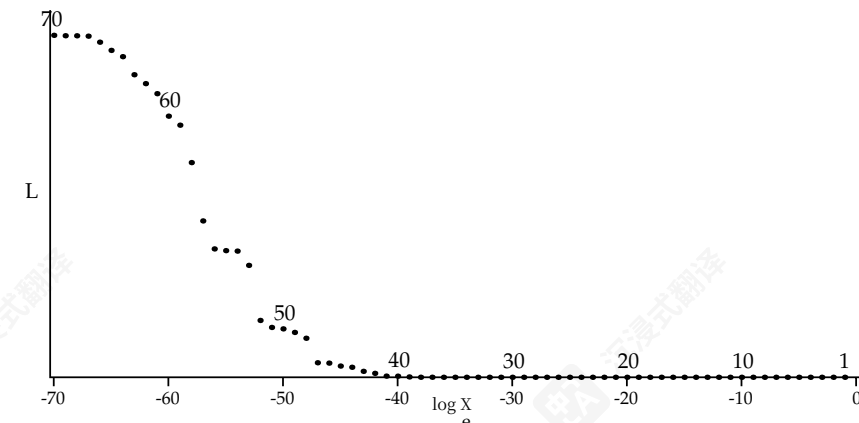


图 7: 与图 6 相同的嵌套采样点序列, 显示在它们 X 值的粗略中心估计处 (即在对数 X 上均匀分布)。

偏差。

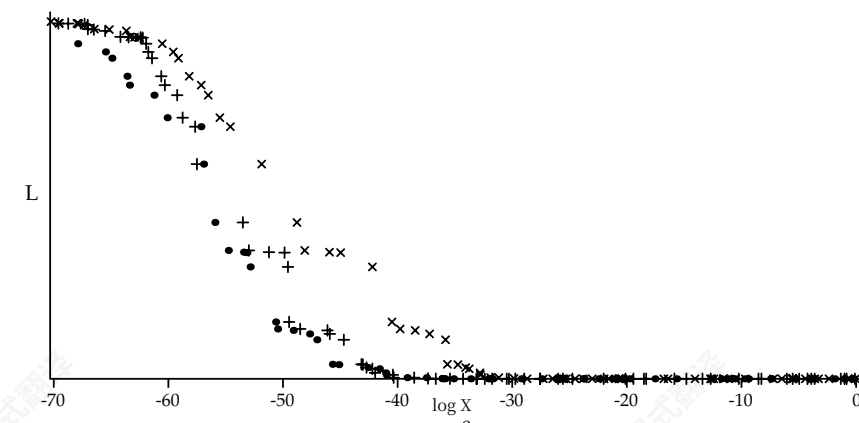


图 8: 与图 6 相同的嵌套采样点序列, 显示三个随机分配 ($\bullet$, $+$, $\times$) 的 X 值, 这些值来自 $\text{Pr}(X)$ 。

通过从它们的 pdf (14) 中采样 (*i.e.* 重复猜测) 质量比 $\mathbf{t}$, 并累积由此产生的 Z 值, 可以获得一个更专业但相当相似的估计 $\log Z = -42.7 \pm 5.5$ (再次是均值和标准差)。图 8 中显示了这三个猜测。正如预期的那样, 曲线的形状大部分相同, 主要差异在于对数 X 中的随机偏移。在多次分配 $\mathbf{t}$ 的平均值上, 这个更好的估计值偏离真实值 $\log Z = -37.81$ 0.9 个标准差, 这并不令人意外。

当然, 需要足够多的迭代 (至少 NH 次) 来达到并越过后验质量的主体部分。似然值可以通过指示 Z 的累积似乎开始出现尾部来指导终止决策, 例如 (17), 但很少有一个明确的准则。

Incidentally, the discerning reader will have noticed that the original dimensionality of the example disappeared early on. The problem could just as well have been one-dimensional, described by (29). Nested sampling ignores dimensionality, off-loading such complications to the task of sampling within the likelihood constraint.

10 Multiple runs

In repeated trials of nested sampling, the 50% interquartile range for Z is observed to cover the truth about half of the time, and similarly for other confidence intervals. Logically, though, that frequentist observation is pointless because the test is forced to succeed. It could only fail if there were an error, either in the logic of nested sampling or in the programming. Nevertheless, it raises the question of how multiple runs should best be combined. Loosely, one might invent some *ad hoc* averaging, but there ought to be a better way.

Suppose that several runs $r = 1, 2, \dots$ are undertaken with numbers $N^{(r)}$ of points, and that run r accumulates likelihood values $L_1^{(r)} < L_2^{(r)} < L_3^{(r)} < \dots$. Merge these values into a single global sequence

$$\mathcal{L}_1 < \mathcal{L}_2 < \mathcal{L}_3 < \dots \quad (31)$$

and consider the status of the runs as they pass any likelihood value L^* . Unless run r has already been terminated, it will at that time have its $N^{(r)}$ points uniformly distributed in X (as always), but subject to their likelihoods being above L^* , so that their enclosed prior masses are leftward of $X^* = X(L^*)$ as illustrated in Figure 9. The run's rightmost point (with worst L) is distributed along X as the largest of $N^{(r)}$ random numbers from $\text{Uniform}(0, X^*)$.

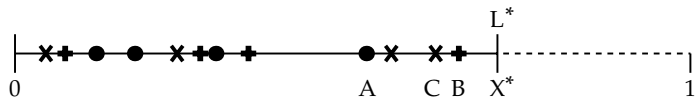


Figure 9: $N = 4$ points each from 3 runs ($\bullet, +, \times$) as they pass L^* . A,B,C are their rightmost points, respectively. B is the rightmost of all 12 points.

Meanwhile, the rightmost point of all (“B” in Figure 9) is being distributed as the largest of all these $\nu = N^{(1)} + N^{(2)} + N^{(3)} + \dots$ random numbers from $\text{Uniform}(0, X^*)$. But, if X^* is taken to be the i 'th element $\mathcal{X}_i$ of likelihood $\mathcal{L}_i$ in the global sequence, then this rightmost of all is the subsequent element $\mathcal{X}_{i+1}$. The shrinkage factor $t_{i+1} = \mathcal{X}_{i+1}/\mathcal{X}_i$ is distributed as

$$\text{Pr}(t) = \nu t^{\nu-1}. \quad (32)$$

Hence the merged combination of individual runs behaves just like a single run with the combined number of points, and can be analyzed as such with no need for any damaging *ad hoc* fixup. Presumably, this is how nested sampling might be conveniently implemented on parallel hardware.

11 Philosophy

Note that our inferences are properly probabilistic. O'Hagan (1987) has criticised Monte Carlo integration for being frequentist, because results depend on the random generator with its

顺便说一句，敏锐的读者可能已经注意到，示例的原始维度很早就消失了。这个问题同样可以是维度的，由 (29) 描述。嵌套抽样忽略了维度，将此类复杂性加载到在似然约束内进行采样的任务中。

10 多次运行

在嵌套抽样的重复试验中，观察到 Z 的 50% 四分位距大约一半时间能覆盖真相，其他置信区间也是如此。然而，从逻辑上讲，这种频率论观察是无意义的，因为测试被强迫成功。它只有在逻辑或编程出现错误时才会失败。尽管如此，它提出了如何最佳组合多次运行的问题。粗略来说，人们可能会发明某种 *ad hoc* 平均法，但应该有更好的方法。

假设进行多次运行 $r = 1, 2, \dots$ ，使用点数 $N^{(r)}$ ，运行 r 累积似然值 $L_1^{(r)} < L_2^{(r)} < L_3^{(r)} < \dots$ 。将这些值合并为单个全局序列 ...

$$\mathcal{L}_1 < \mathcal{L}_2 < \mathcal{L}_3 < \dots \quad (31)$$

并考虑运行在通过任何似然值 L^* 时的状态。除非运行 r 已经终止，否则在该时刻，其 $N^{(r)}$ 点将均匀分布在 X 中（一如既往），但受其似然值高于 L^* 的约束，因此其包含的先验质量位于 $X^* = X(L^*)$ 左侧，如图 9 所示。该运行的最右点（具有最差 L ）沿着 X 分布，作为来自 $\text{Uniform}(0, X^*)$ 的最大 $N^{(r)}$ 个随机数的分布。

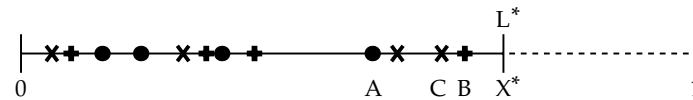


图9：从3次运行 ($\bullet, +, \times$) 中各取 $N = 4$ 个点，随着它们通过 L^* 。A、B、C 分别是它们最右侧的点，依次排列。B 是所有 12 个点中最右侧的点。

与此同时，所有点中最右侧的点（图9中的“B”）被分配为所有这些来自均匀分布 $\text{Uniform}(0, X^*)$ 的 $\nu = N^{(1)} + N^{(2)} + N^{(3)} + \dots$ 个随机数中的最大值。但是，如果将 X^* 取为全局序列中第 i 个元素 $\mathcal{X}_i$ 的似然度，那么这个所有点中最右侧的点就是后续元素 $\mathcal{X}_{i+1}$ 。收缩因子 $t_{i+1} = \mathcal{X}_{i+1}/\mathcal{X}_i$ 的分布为

$$\text{Pr}(t) = \nu t^{\nu-1}. \quad (32)$$

因此，单个运行合并的组合行为就像一个具有组合点数的单个运行一样，可以像分析单个运行那样进行分析，而无需任何破坏性的 fixup。据说，这就是嵌套抽样如何在并行硬件上方便实现的方式。

11 哲学

请注意，我们的推断是恰当的概率论方法。O'Hagan (1987) 批评蒙特卡洛积分是频率派的，因为结果取决于随机生成器及其

sampling distribution, as well as on the likelihood function. This criticism has entered the folklore and makes it difficult for Bayesians, who have been using frequentist algorithms to generate their results, to argue convincingly against frequentist methodology. Fortunately, the criticism does not apply to nested sampling.

It is true that our results depend on the generator which, by sampling θ within the likelihood constraint, thereby implicitly samples the shrinkage factors t according to (14). Yet nested sampling takes that into account. The numerical integration rule (whether trapezoid or an alternative) imposes a definitive functional form $Z(\mathbf{t})$ on the evidence values, parameterised by the evaluated likelihoods. Accordingly, the known $\text{Pr}(\mathbf{t})$ induces a correspondingly unambiguous $\text{Pr}(Z)$ through the multi-dimensional integral (19).

In the analogy with ordinary Bayesian inference from data, the shrinkage factors $\mathbf{t}$ play the role of noise of known distribution, the algorithm plays the role of observing equipment, and the likelihood evaluations play data from which we infer $\text{Pr}(Z \mid \text{data})$. Implicit in this is that we knew nothing else of significance about Z or its underlying likelihood function. If we did have some such expectation (Rasmussen and Gharamani (2003)), we could factor those probabilities into the calculation, and thereby improve the results. In general it seems unlikely that there would be much to gain, but there might be cases where something useful was known, and the facility is present.

Note that integrating the monotonic function $L(X)$ is quite different from integrating the arbitrary $f(x)$ that O’Hagan had in mind. Suppose, to take a limiting illustration, that we knew $f(0) = f(1) = c$. If f were arbitrary, the uncertainty in its integral would be substantial, and questionable. O’Hagan made that point in greater generality. If f is monotonic, though, it has to be constant, and $\int_0^1 f(x)dx = c$, with no uncertainty. Nested sampling generalises this without any compromise with the ordinary rules of probability theory.

O’Hagan also criticised the Monte Carlo integration of his day for producing results which depended on the sampling distribution. Thus, if the likelihood function factorises as $L = L_1 L_2$, then the result $Z = \int (L_1 L_2) dX$ of sampling over X will differ from the result $Z' = \int L_1 (L_2 dX)$ of including L_2 with X and sampling from that combination instead. But that’s not a criticism of nested sampling; it’s a straightforward feature, to be expected and approved. If we are clever enough to factor part of the likelihood into the prior, and sample from that, we would thereby start closer to the posterior, and can expect to be rewarded with a better estimate having diminished uncertainty. Conversely, if we were foolish enough to retreat away from the posterior by dividing some factor out of the prior, then we should expect to pay for it with increased uncertainty. This is just how a properly constituted algorithm for inference ought to behave. Anything else would be worrisome.

In summary, nested sampling obeys the rules of probability calculus. Accordingly, and as befits an algorithm for Bayesian computation, its nature is Bayesian, not frequentist.

12 Density of states

The density of states (being the prior mass in a thin likelihood shell — loosely, its area) is often defined with respect to “energy” $E = -\log L$ as $g = dX/dE = -dX/d \log L$, but here it is more convenient to define it in fully logarithmic form as

$$g^*(L) = -\frac{d \log X}{d \log L}. \quad (33)$$

抽样分布，以及似然函数。这种批评已经流传甚广，使得使用频率派算法生成结果的贝叶斯学者难以有力地反驳频率派方法。幸运的是，这种批评不适用于嵌套抽样。

确实，我们的结果取决于生成器，该生成器通过在似然约束内采样 θ ，从而隐式地采样收缩因子 t （根据公式14）。然而嵌套采样会考虑这一点。数值积分规则（无论是梯形法则还是其他方法）对证据值施加了明确的函数形式 $Z(\mathbf{t})$ ，其特征是评估的似然值。因此，已知的 $\text{Pr}(\mathbf{t})$ 通过多维积分（公式19）诱导出相应的不明确 $\text{Pr}(Z)$ 。

在与普通贝叶斯数据推断的类比中，收缩因子 $\mathbf{t}$ 扮演已知分布的噪声的角色，算法扮演观测设备的角色，而似然评估则扮演我们从其中推断 $\text{Pr}(Z \mid \text{data})$ 的数据。这其中隐含的是我们对 Z 或其潜在似然函数没有任何其他重要了解。如果我们确实有一些这样的预期（Rasmussen和Gharamani (2003)），我们可以将这些概率纳入计算，从而改进结果。一般来说，似乎不太可能获得太多收益，但可能存在一些已知某些有用信息且该功能可用的情况。

请注意，将单调函数 $L(X)$ 积分与 O’Hagan 想到的任意 $f(x)$ 积分有相当大的不同。假设，为了说明极限情况，我们知道 $f(0) = f(1) = c$ 。如果 f 是任意的，那么它的积分中的不确定性将是巨大的，并且是有问题的。O’Hagan 更普遍地指出了这一点。然而，如果 f 是单调的，那么它必须是常数，并且 $\int_0^1 f(x)dx = c$ ，没有不确定性。嵌套采样推广了这一点，而没有对概率论的普通规则做出任何妥协。

O’Hagan 还批评了他那个时代的蒙特卡洛积分，因为它产生的结果依赖于抽样分布。因此，如果似然函数分解为 $L = L_1 L_2$ ，那么在 X 上抽样得到的结果 $Z = \int (L_1 L_2) dX$ 将与包含 L_2 和 X 并从该组合中抽样的结果 $Z' = \int L_1 (L_2 dX)$ 不同。但这并不是对嵌套抽样的批评；它是一个直接的特征，可以预期并得到认可。如果我们足够聪明，将似然的一部分分解到先验中，并从该部分抽样，那么我们就更接近后验，并有望得到不确定性降低的更好估计。相反，如果我们愚蠢到通过从先验中除掉某个因子而远离后验，那么我们应该预期为此付出不确定性增加的代价。这正是推理算法应该表现的方式。任何其他情况都令人担忧。

总而言之，嵌套采样遵循概率演算的规则。因此，作为贝叶斯计算算法，其本质是贝叶斯而非频率派的。

12 状态密度

态密度（即在一个薄似然壳中的先验质量，大致上，其面积）通常相对于“能量” $E = -\log L$ 定义为 $g = dX/dE = -dX/d \log L$ ，但在这里，将其以完全对数形式定义更为方便，即

$$g^*(L) = -\frac{d \log X}{d \log L}. \quad (33)$$

Differencing across r steps gives

$$g^*(\bar{L}) \leftarrow -\frac{\log X_i - \log X_{i-r}}{\log L_i - \log L_{i-r}} = \frac{-\log t_i - \log t_{i-1} - \dots - \log t_{i-r+1}}{\log L_i - \log L_{i-r}} \quad (34)$$

for $\bar{L}$ somewhere between L_{i-r} and L_i . The statistics (15) of each $\log t$ are known, and independent, so that in terms of mean and standard deviation

$$g^* \leftarrow \frac{(r \pm \sqrt{r})/N}{\log L_i - \log L_{i-r}}. \quad (35)$$

As usual in numerical differentiation, the formal uncertainty diminishes as the chosen interval widens, but the difference ratio relates less precisely to the required differential.

Individual steps ($r = 1$) estimate g^* with 100% expected error. Even so, these steps underlie the evidence summation and are the most basic results of the computation. Individual steps can also build properties other than the evidence (known in thermodynamics as the partition function). In particular, the annealed partition function

$$Z(\beta) = \int L^\beta dX \quad (36)$$

is available at any inverse temperature β , provided the computation is carried far enough inward to cover the bulk of the required integral. Nested sampling is not thermal, but can simulate any temperature.

13 Posterior

Representative samples $\tilde{\theta}$ from the posterior density are defined by sampling from the posterior distribution $p(\theta)$, which is simply the prior weighted by likelihood. Equivalently, they can be obtained by sampling randomly from the area Z under the one-dimensional curve $L(X)$, as shown in Figure 10.

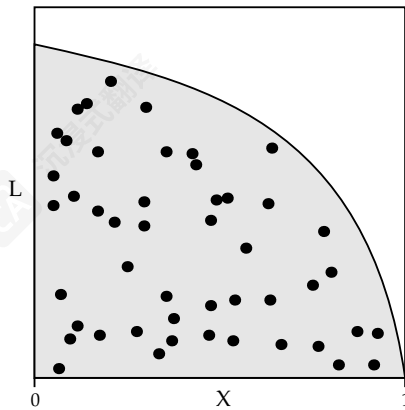


Figure 10: Posterior samples are scattered randomly over the area Z .

Because this area Z is already decomposed (8) into $\sum_i L_i w_i$, we can select our samples by choosing from these sub-areas. In other words, the existing sequence of points $\theta_1, \theta_2, \theta_3, \dots$

在 r 步之间求导给出

$$g^*(\bar{L}) \leftarrow -\frac{\log X_i - \log X_{i-r}}{\log L_i - \log L_{i-r}} = \frac{-\log t_i - \log t_{i-1} - \dots - \log t_{i-r+1}}{\log L_i - \log L_{i-r}} \quad (34)$$

用于 $\bar{L}$ 介于 L_{i-r} 和 L_i 之间。每个 $\log t$ 的统计数据 (15) 是已知的，且相互独立，因此在均值和标准差方面

$$g^* \leftarrow \frac{(r \pm \sqrt{r})/N}{\log L_i - \log L_{i-r}}. \quad (35)$$

与数值微分中的惯例一样，随着所选区间的加宽，形式不确定性会减小，但差异比与所需的微分关系不那么精确。

单个步骤 ($r = 1$) 以 100% 的预期误差估计 g^* 。即便如此，这些步骤构成了证据求和的基础，并且是计算中最基本的结果。单个步骤还可以构建除证据之外的其他属性（在热力学中称为配分函数）。特别是退火配分函数

$$Z(\beta) = \int L^\beta dX \quad (36)$$

在任何倒数温度 β 下都可用，前提是计算足够深入内部以覆盖所需积分的大部分。嵌套抽样不是热力学过程，但可以模拟任何温度。

13 后验

后验密度中的代表性样本 θ 由从后验分布 $p(\theta)$ 中采样定义，后验分布只是先验分布与似然度的加权。等效地，它们可以通过从一维曲线 $L(X)$ 下的区域 Z 中随机采样获得，如图 10 所示。

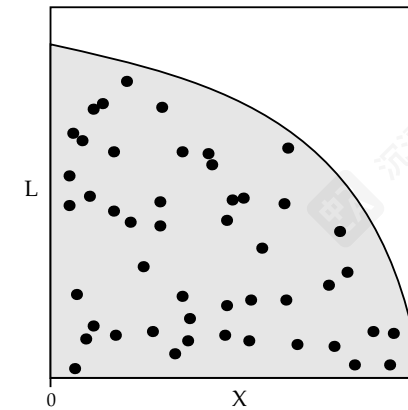


图 10：后验样本随机散布在区域 Z 上。

由于该区域 Z 已被分解 (8) 为 $\sum_i L_i w_i$ ，我们可以通过从中选择子区域来选取样本。换句话说，现有的点序列 $\theta_1, \theta_2, \theta_3, \dots$

already gives a set of posterior representatives, provided the i 'th is assigned the appropriate importance weight $L_i w_i$, normalised by Z to yield a probability with unit total. For a given choice of coefficients $\mathbf{t}$, the posterior probability for point i would be

$$p_i(\mathbf{t}) = L_i w_i(\mathbf{t}) / Z(\mathbf{t}). \quad (37)$$

Taking into account the uncertainty of the coefficients, this posterior probability for point i becomes

$$p_i \longleftarrow \int \frac{L_i w_i(\mathbf{t})}{Z(\mathbf{t})} \Pr(\mathbf{t}) d\mathbf{t} = L_i \left\langle \frac{w_i(\mathbf{t})}{Z(\mathbf{t})} \right\rangle_{\mathbf{t}}, \quad (38)$$

which can be evaluated by Monte Carlo as before.

To obtain equally-weighted posterior representatives, all we need do is accept point θ_i with probability p_i/K , where $K \geq \max_j(p_j)$ must be large enough to avoid duplication. The maximum number of representative posterior samples is given by the entropy (*i.e.* the Shannon channel capacity) as

$$\mathcal{N} = \exp \left(- \sum_{i=1}^m p_i \log p_i \right), \quad (39)$$

though in practice the available number is somewhat less because of the maximisation limit K . For the approximately rank- C multivariate normal likelihood conjectured earlier, we may expect $\mathcal{N} \sim N\sqrt{C}$. More precisely, we saw after (11) that this form of likelihood would have a posterior of standard deviation $\sqrt{C/2}$ with respect to $\log X$. The step number i is expected to be roughly $-N \log X$, so p_i should be roughly Gaussian in i , with standard deviation $N\sqrt{C/2}$. In that case, (39) evaluates to $\mathcal{N} = N\sqrt{\pi e C}$. Hence we may assign

$$C \longleftarrow \mathcal{N}^2 / \pi e N^2 \quad (40)$$

as our estimate of the effective rank (number of useful principal components) of the likelihood. It might be nice to know this, and the diagnostic can easily be accumulated along with Z and H .

14 Quantification

Suppose we wish to quantify property $Q(\theta)$, for which point θ_i carries value $Q_i = Q(\theta_i)$. For this, we seek the posterior distribution $\Pr(Q)$. This could be estimated from the posterior samples just obtained, through representative values $Q(\tilde{\theta})$. More accurately, we can use the full sequence of nested points directly, without reducing it to a less informative equally-weighted subset. Each value Q_i is associated with probabilistic weight $p_i(\mathbf{t})$ from (37). In particular, the mean and standard deviation of Q (if they exist) are as usual obtainable from the first and second moments

$$\begin{aligned} \mu(\mathbf{t}) &= \sum_{i=1}^m Q_i p_i(\mathbf{t}) \\ \sigma(\mathbf{t}) &= \left(\sum_{i=1}^m Q_i^2 p_i(\mathbf{t}) - \mu(\mathbf{t})^2 \right)^{1/2} \end{aligned} \quad (41)$$

for any specified sequence $\mathbf{t}$. Again, we ought to acknowledge uncertainty by invoking the sequence probability. Just as for the evidence in (20), the spread over $\mathbf{t}$ yields mean and deviation

$$\begin{aligned} \mathbf{E}(Q) &\longleftarrow \text{estimate} \pm \text{uncertainty, from } \{\mu\}_{\mathbf{t}} \\ \text{dev}(Q) &\longleftarrow \text{estimate} \pm \text{uncertainty, from } \{\sigma\}_{\mathbf{t}}. \end{aligned} \quad (42)$$

As was the case for Z , these numerical uncertainties are caused by our limited sequence of nested points, not by any small errors in their evaluation, whether by Monte Carlo or otherwise. With

已经给出了一组后验代表, 前提是 i 被分配适当的重要性权重 $L_i w_i$, 并通过 Z 归一化以产生具有单位总和的概率。对于给定的系数 $\mathbf{t}$ 选择, 点 i 的后验概率将是

$$p_i(\mathbf{t}) = L_i w_i(\mathbf{t}) / Z(\mathbf{t}). \quad (37)$$

考虑到系数的不确定性, 点 i 的后验概率变为

$$p_i \longleftarrow \int \frac{L_i w_i(\mathbf{t})}{Z(\mathbf{t})} \Pr(\mathbf{t}) d\mathbf{t} = L_i \left\langle \frac{w_i(\mathbf{t})}{Z(\mathbf{t})} \right\rangle_{\mathbf{t}}, \quad (38)$$

这可以通过蒙特卡洛方法像之前一样进行评估。

为了获得等权重的后验代表, 我们只需要以概率 p_i/K 接受点 θ_i , 其中 $K \geq \max_j(p_j)$ 必须足够大以避免重复。后验代表的最大数量由熵 (即香农信道容量) 给出, 为

$$\mathcal{N} = \exp \left(- \sum_{i=1}^m p_i \log p_i \right), \quad (39)$$

尽管由于最大化限制 K , 实际可用的数量会稍小。对于之前假设的大约秩- C 多元正态似然, 我们可能期望 $\mathcal{N} \sim N\sqrt{C}$ 。更精确地说, 我们在 (11) 之后看到, 这种形式的似然的后验标准差为 $\sqrt{C/2}$, 关于 $\log X$ 。步数 i 预计约为 $-N \log X$, 因此 p_i 在 i 中应该大致呈高斯分布, 标准差为 $N\sqrt{C/2}$ 。在这种情况下, (39) 评估为 $\mathcal{N} = N\sqrt{\pi e C}$ 。因此我们可以赋值

$$C \longleftarrow \mathcal{N}^2 / \pi e N^2 \quad (40)$$

作为我们对似然有效秩 (有用主成分数量) 的估计。知道这一点可能很有用, 并且诊断结果可以很容易地与 Z 和 H 一起累积。

14 量化

假设我们希望量化属性 $Q(\theta)$, 其中点 θ_i 具有值 $Q_i = Q(\theta_i)$ 。为此, 我们寻求后验分布 $\Pr(Q)$ 。这可以从刚刚获得的后验样本中估计, 通过代表性值 $Q(\tilde{\theta})$ 。更准确地说, 我们可以直接使用嵌套点的完整序列, 而无需将其减少为信息量较少的等权重子集。每个值 Q_i 都与来自 (37) 的概率权重 $p_i(\mathbf{t})$ 相关。特别是, 如果它们存在, Q 的均值和标准差通常可以从 $\mathbf{t}$ 首要和第二矩中获得

$$\begin{aligned} \mu(\mathbf{t}) &= \sum_{i=1}^m Q_i p_i(\mathbf{t}) \\ \sigma(\mathbf{t}) &= \left(\sum_{i=1}^m Q_i^2 p_i(\mathbf{t}) - \mu(\mathbf{t})^2 \right)^{1/2} \end{aligned} \quad (41)$$

对于任何特定的序列 $\mathbf{t}$ 。同样, 我们应该通过调用序列概率来承认不确定性。正如 (20) 中的证据一样, $\mathbf{t}$ 的分布产生均值和偏差

$$\begin{aligned} \mathbf{E}(Q) &\longleftarrow \text{estimate} \pm \text{uncertainty, from } \{\mu\}_{\mathbf{t}} \\ \text{dev}(Q) &\longleftarrow \text{estimate} \pm \text{uncertainty, from } \{\sigma\}_{\mathbf{t}}. \end{aligned} \quad (42)$$

对于 Z 来说, 这些数值不确定性是由我们嵌套点的序列有限性引起的, 而不是由它们评估中的任何小误差 (无论是蒙特卡洛方法还是其他方法) 造成的。有了

sufficient resources, we could increase N to make these uncertainties arbitrarily small, but of course we would not thereby eliminate $\text{dev}(Q)$, which is caused by the likelihood failing to specify θ precisely, and is part of what we ought to want to know about Q . However, if the numerical uncertainty on the mean $\mathbb{E}(Q)$ exceeded the standard deviation $\text{dev}(Q)$, one might question the utility of the computation, and wish to repeat it more slowly with more points.

15 Annealing

Nested sampling is related to simulated annealing, which uses fractional powers L^β of the likelihood to move gradually from the prior ($\beta = 0$) to the posterior ($\beta = 1$). As the inverse temperature β increases, annealing softly compresses points $\{\theta\}$ sampled from $dP_\beta \propto L^\beta dX$, known as a thermalised ensemble. At stage β , the mean log-likelihood

$$\langle \log L \rangle_\beta = \int \log L \, dP_\beta = \frac{\int L^\beta \log L \, dX}{\int L^\beta dX} = \frac{d}{d\beta} \log \int L^\beta dX \quad (43)$$

is estimated by averaging over the corresponding ensemble. Summing this yields

$$\int_0^1 \langle \log L \rangle_\beta \, d\beta = \log \int L \, dX - \log \int dX = \log Z, \quad (44)$$

which is the thermodynamic integration formula. It is not normally accompanied by any estimate of uncertainty, presumably because the uncertainty in $\langle \log L \rangle$ is difficult to assess.

The bulk of the ensemble, with respect to $\log X$, should follow the posterior $dP_\beta \propto L^\beta X \, d \log X$ and be found around the maximum of $L^\beta X$. Under the usual conditions of differentiability and concavity “ $\frown$ ”, this maximum occurs where

$$g^* = -\frac{d \log X}{d \log L} = \beta. \quad (45)$$

Annealing over β thus tracks the density-of-states g^* , equivalent to $-1/\text{slope}$ on a $\log L / \log X$ plot, whereas nested sampling tracks the underlying abscissa value $\log X$.

As β increases from 0 to 1, one hopes that the annealing maximum tracks steadily up in L , so inward in X (Figure 11a). The annealing schedule that dictates how fast β is increased ought to allow successive posteriors P_β to overlap substantially — exactly how much is still a matter of some controversy. Yet it may not be possible at all.

Suppose that g^* is not an increasing function of $\log X$, so that $L^\beta X$ is not concave (Figure 11b). No matter what schedule is adopted, annealing is supposed to follow the concave hull of the log-likelihood function as its tangential slope flattens. But this will require jumping right across any convex “ $\smile$ ” region that separates ordinary concave “phases” where local maxima of $L^\beta X$ are to be found. At $\beta = 1$, the bulk of the posterior should lie near a maximum of LX , in one or other of these phases. Let us call the outer phase “steam” and the inner phase “water”, as suggested by the potentially large difference in volume. Annealing to $\beta = 1$ will normally take the ensemble from the neighbourhood of A to the neighbourhood of B, where the slope is $d \log L / d \log X = -1/\beta = -1$. Yet we actually want samples to be found from the inner phase beyond D, finding which will be exponentially improbable unless the intervening convex valley is shallow. Alternatively, annealing could be taken beyond $\beta = 1$ until, when the ensemble is near the point of inflection C, the supercooled steam crashes inward to chilled water, somewhere near F. It might then be possible to anneal back out

足够的资源，我们可以增加 N 使这些不确定性任意小，但当然我们不会因此消除 $\text{dev}(Q)$ ，它是由似然函数未能精确指定 θ 引起的，并且是我们应该了解的关于 Q 的一部分。然而，如果均值 $\mathbb{E}(Q)$ 的数值不确定性超过了标准偏差 $\text{dev}(Q)$ ，人们可能会质疑计算的效用，并希望用更多点更慢地重复它。

15 退火

嵌套采样与模拟退火相关，后者使用似然函数的分数幂 L^β 从先验 ($\beta = 0$) 逐渐移动到后验 ($\beta = 1$)。随着逆温度 β 的增加，退火会柔和地压缩从 $dP_\beta \propto L^\beta dX$ 抽取的点 $\{\theta\}$ ，这些点被称为热平衡集合。在阶段 β ，平均对数似然

$$\langle \log L \rangle_\beta = \int \log L \, dP_\beta = \frac{\int L^\beta \log L \, dX}{\int L^\beta dX} = \frac{d}{d\beta} \log \int L^\beta dX \quad (43)$$

通过对应集合上的平均来估计。求和后得到

$$\int_0^1 \langle \log L \rangle_\beta \, d\beta = \log \int L \, dX - \log \int dX = \log Z, \quad (44)$$

这是热力学积分公式。它通常不伴随任何不确定性估计，推测是因为〈对数 L 〉的不确定性难以评估。

对于对数 $L^\beta X$ ，集合的大部分应该遵循后验对数，并出现在 $L^\beta X$ 的最大值附近。在可微性和凹性 $\frown$ 的通常条件下，这个最大值出现在

$$g^* = -\frac{d \log X}{d \log L} = \beta. \quad (45)$$

退火过程跨越 β 从而追踪态密度 g^* ，相当于 $-1/\text{对数 } L / \text{对数 } X$ 图上的斜率，而嵌套采样则追踪底部的横坐标值对数 X 。

当 β 从 0 增加到 1 时，人们希望退火最大值在 L 中稳步上升，在 X 中向内（图 11a）。决定 β 如何快速增加的退火调度应当允许连续的后验分布 P_β 显著重叠，具体多少仍然存在一些争议。然而，这或许根本就不可能实现。

假设 g^* 不是 $\log X$ 的增函数，因此 $L^\beta X$ 不是凹的（图 11b）。无论采用何种计划，退火过程都应沿着对数似然函数的凹包络线进行，其切线斜率 g 会逐渐变平。但这将需要跨越任何将普通凹相（其中可以找到 $L^\beta X$ 的局部最大值）分隔开凸的 “ $\smile$ ” 区域。在 $\beta = 1$ ，后验分布的大部分应该位于 LX 的一个或另一个相的最大值附近。让我们将外相称为“蒸汽”，内相称为“水”，正如体积可能存在巨大差异所暗示的那样。退火到 $\beta = 1$ 通常会将集合从 A 附近移动到 B 附近，此时斜率为 $d \log L / d \log X = -1/\beta = -1$ 。然而，我们实际上希望从内相 D 以外的区域找到样本，而找到这些样本将呈指数级不可能，除非中间的凸形山谷很浅。或者，退火可以继续进行到 $\beta = 1$ ，直到当集合接近拐点 C 时，过冷蒸汽会向内崩溃到冷却的水中，大约在 F 处。那时，可能有可能退火回

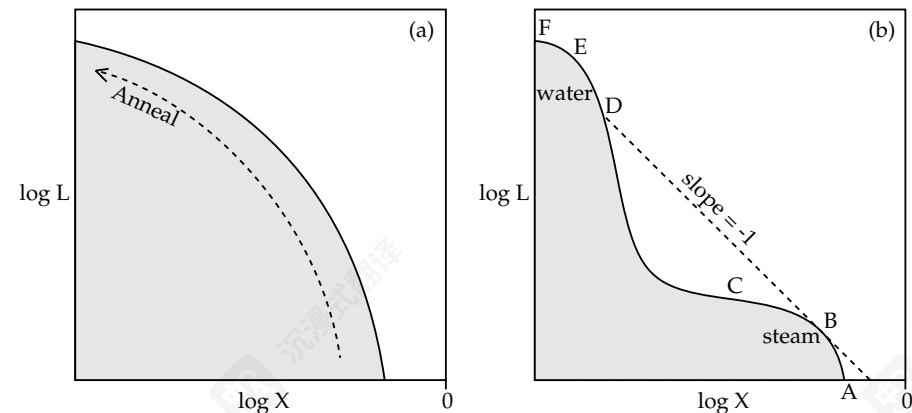


Figure 11: Proper annealing needs log-likelihood to be concave like (a), not (b).

to unit temperature, reaching the desired water phase near E. However, annealing no longer bridges smoothly during the crash, and the value of the evidence is lost. Lost along with it is the internal Bayes factor $\Pr(\text{states near E})/\Pr(\text{states near B})$, which might have enabled the program to assess the relative importance of water and steam. If there were three phases instead of just two, annealing might fail even more spectacularly. It would be quite possible for supercooled steam to condense directly to cold ice, and superheated ice to sublime directly to hot steam, without settling in an intermediate water phase at all. The dominant phase could be lost in the hysteresis, and inaccessible to annealing.

Phase change problems in general are well known to be difficult to anneal, and especially so when of first order as here. Nested sampling, though, marches steadily down in prior mass X along ABCDEF..., regardless of whether the associated log-likelihood is concave or convex or even differentiable at all. There is no analogue of temperature, so there is never any thermal catastrophe. Nested points will pass through the steam phase to the supercooled region, then steadily into superheated water until the ordinary water phase is reached, traversed, and left behind in an optional continued search for ice. All the internal Bayes factors are available, so the dominant phase can be identified and quantified.

16 Physics example

For an example from physics, consider the following elementary model of order/disorder. Atoms can be in either of two states, 0 or 1. A sequence of n atoms is laid out along a line, so there are 2^n equally-weighted prior states. The atoms define a sequence of clusters c with widths h_c across which the state is constant. For example, the ten atoms 0001111001 have four clusters of width $h_1 = 3$ (zeros), $h_2 = 4$ (ones), $h_3 = 2$ (zeros), $h_4 = 1$ (one). Each cluster has an energy benefit (*i.e.* a log-likelihood gain) proportional to the number $\frac{1}{2}h(h-1)$ of internal interactions permitted among its members, so that (with specific scaling)

$$\log L = (2/n) \sum_c \frac{1}{2} h_c (h_c - 1), \quad \sum_c h_c = n. \quad (46)$$

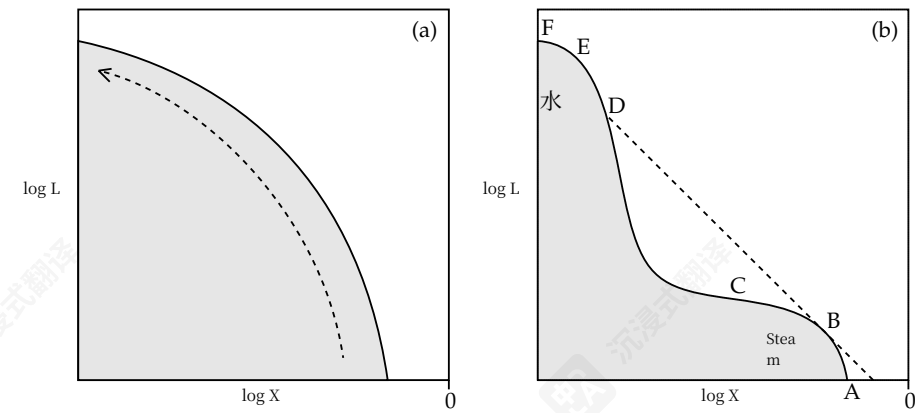


图11: 正确的退火需要对数似然函数像(a)一样呈凹形, 而不是(b)的样子。

到单位温度, 在 E 附近达到所需的水相。然而, 在崩溃过程中退火不再平滑地连接, 证据的值就会丢失。随着它一起丢失的是内部贝叶斯因子 $\Pr(\text{靠近 E 的状态})/\Pr(\text{靠近 B 的状态})$, 这可能使程序能够评估水和蒸汽的相对重要性。如果有三个相而不是两个, 退火可能会更加壮观地失败。超冷蒸汽可能会直接凝结成冷水冰, 而超热冰可能会直接升华成热蒸汽, 而根本不会在中间的水相中停留。主导相可能会在滞后现象中丢失, 并且无法通过退火访问。

相变问题通常众所周知难以退火, 尤其是当像这里这样是一阶相变时。然而, 嵌套采样会稳步地沿着先验质量 ABCDEF... 下降, 无论相关的对数似然函数是凹的还是凸的, 甚至是否可微。没有温度的类似物, 所以永远不会发生热灾难。嵌套点会通过蒸汽相进入过冷区域, 然后稳步进入过热水, 直到达到普通水相, 穿过并留在可选的继续寻找冰的搜索中。所有内部贝叶斯因子都可用, 因此主导相可以被识别和量化。

16 物理示例

以物理学为例, 考虑以下关于有序/无序的简单模型。原子可以处于两种状态之一, 0 或 1。一个由 n 个原子组成的序列沿直线排列, 因此有 2^n 个等权重的先验状态。这些原子定义了一个由 c 个簇组成的序列, 这些簇的宽度为 h_c , 在它们内部状态保持不变。例如, 十个原子 0001111001 有四个宽度为 $h_1 = 3$ (0)、 $h_2 = 4$ (1)、 $h_3 = 2$ (0)、 $h_4 = 1$ (1) 的簇。每个簇都有一个与其成员之间允许的内部交互数量 $\frac{1}{2}h(h-1)$ 成正比的能量收益 (*i.e.* 对数似然增益), 因此 (在特定的缩放下)

$$\log L = (2/n) \sum_c \frac{1}{2} h_c (h_c - 1), \quad \sum_c h_c = n. \quad (46)$$

Of the 1024 states of 10 atoms, the top two (0000000000 and 1111111111) were fully ordered with $\log L = 9$ and shared 49% of the posterior, the next four (0000000001, 0111111111, 1000000000, 1111111110) with $\log L = 7.2$ shared another 16%, the example ten atoms 0001111001 had $\log L = 2$, and so on down to the two lowest states (0101010101 and 1010101010) with $\log L = 0$, which shared 0.006%.

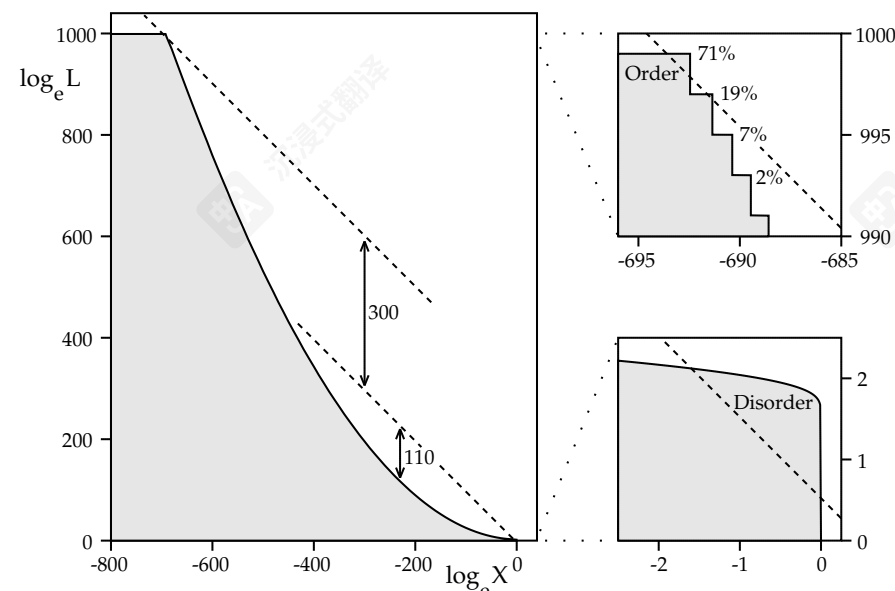


Figure 12: Order/disorder example for 1000 atoms. The upper sub-plot magnifies the “order” phase, and the lower sub-plot magnifies the “disorder” phase. The order phase is favoured by a Bayes factor $\exp(300)$, but is hard to find by a factor $\exp(110)$. Dashed lines enclose 75% of posterior samples for each phase.

Figure 12 shows the behaviour for $n = 1000$, precisely calculated by recurrence on n . Again, the “order” phase with wide clusters dominates, with the two fully-ordered states 0000... and 1111... with $\log L = 999$ sharing 71% of the posterior, the next four with $\log L = 997.002$ sharing 19%, and so on. With n being large, the “disorder” phase with most clusters narrow is well separated, but the “order” phase is favoured overall by a Bayes factor of e^{300} .

An ensemble annealed to $\beta = 1$, though, has no chance (technically, about e^{-110} chance) of finding the tiny volume occupied by the “order” states. It ought to transition to the ordered phase at the freezing point $\beta = 0.69$, where the two phases ought to become equally populated, but it won't. As expected, the author's simulation (which started with a random state and evolved by inverting atoms at random according to the usual Metropolis detailed balance) failed to move away from the “disorder” phase in an allotted trillion trial inversions. Even if by incredible luck it had succeeded in finding the “order” phase, it could not have determined the order/disorder Bayes factor, or the evidence $Z = e^{306.8878}$.

Yet nested sampling (with a fresh sample within the likelihood constraint approximated by allowing trial MCMC inversions of each atom ten times per iterate) successfully estimated $\log L$ as a function of $\log X$ and reached the fully-ordered states steadily, in the expected

在 10 个原子的 1024 个状态中，前两个 (0000000000 和 1111111111) 是完全有序的，对数

$L = 9$ 49 0000000001, 0111111111, 共享了 %
 $L = 7.2$ 16
 1000000000, 1111111110 的后验概率，接下来的四个 () 又共享了另一个 %，示例中的十个原子 0001111001 对数为 $L = 2$ ，等等，直到两个最低状态 (0101010101 和 1010101010) 对数为 $L = 0$ ，它们共享了 0.006%。

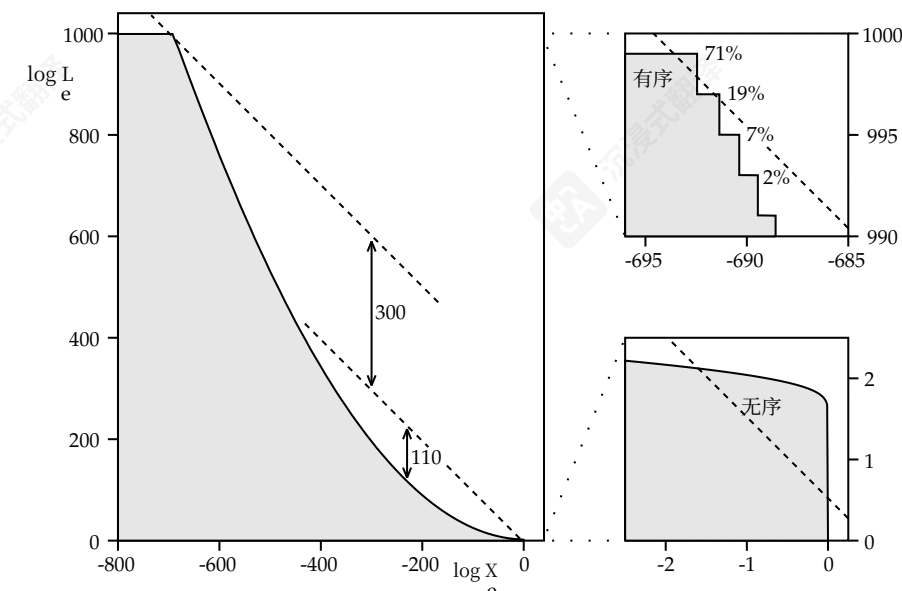


图12: 1000个原子的有序/无序示例。上子图放大了“有序”相，下子图放大了“无序”相。有序相受贝叶斯因子 $\exp(300)$ 的青睐，但难以通过因子 $\exp(110)$ 找到。虚线包围了每个相的75%后验样本。

图12显示了 $n = 1000$ 的行为，这是通过递归在 n 上精确计算的。同样，具有宽集群的“有序”阶段占主导地位，其中两个完全有序的状态 0000... 和 1111... 以 $\log L = 999$ 共享 71% 的后验概率，接下来的四个以 $\log L = 997.002$ 共享 19%，等等。由于 n 很大，“无序”阶段具有大多数窄集群很好地分离，但“有序”阶段总体上以贝叶斯因子 e^{300} 占优势。

一个退火到 $\beta = 1$ 的集合，但（从技术上讲，大约有 e^{-110} 的机会）找不到“有序”状态所占的微小体积。它应该在冻结点 $\beta = 0.69$ 过渡到有序阶段，在那里两个阶段应该变得同样多，但它不会。正如预期的那样，作者的模拟（它从一个随机状态开始，并通过通常的 Metropolis 详细平衡随机反转原子来进化）在分配的一万亿次反转试验中没有成功摆脱“无序”阶段。即使由于难以置信的好运它成功找到了“有序”阶段，它也无法确定有序/无序贝叶斯因子，或证据 $Z = e^{306.8878}$ 。

然而，嵌套采样（在似然约束内使用新鲜样本，通过允许每次迭代对每个原子进行十次试验 MCMC 反演来近似）成功地估计了 $\log L$ 作为 $\log X$ 的函数，并稳步达到完全有序的状态，符合预期

$NH \approx 700N$ iterates.

17 Data analysis example

It is not just large problems with awkward likelihood functions that exhibit phase changes. For an example in data analysis, consider a small experiment to measure the single coordinate θ , over which the prior $\pi(\theta)$ is flat in $(0,1)$. Its data D yield the likelihood function

$$L(\theta) = 0.99 q^{-1} e^{-\theta/q} + 0.01 \quad \text{with, say, } q = 10^{-9}. \quad (47)$$

This is already a decreasing function, and the sorting operation of nested sampling is just the identity, $X = \theta$.

An interpretation of (47) is that the experiment was anticipated to work with 99% reliability. If it worked, the likelihood $L = q^{-1} \exp(-\theta/q)$ would have been appropriate, meaning that $\theta \approx 10^{-9}$ was measured. If it failed, which was anticipated 1% of the time, the likelihood would have been the uninformative $L = 1$, because the equipment would just return a random result. Under annealing, the original hot phase is the failure mode. An annealed ensemble limited to $\beta \leq 1$ is most unlikely to find the “working” mode unless it is allowed millions of trials, and will wrongly suggest “failure”, with $Z = 0.01$. Only if β is increased far beyond 1 to something above e^{1000} would the ensemble be likely to find the working mode in fewer than millions of trials. Even then, the samples would crash inward and have to be annealed back out through those thousand orders of magnitude. And the evidence value would have been lost.

For nested sampling, which steadily tracks $\log X$ instead of trying to use the slope, these problems are easy. All one needs is the determination to keep going for the NH or so shrinkage steps needed to reach and then cross the dominant mode with a collection of N points. By then, the behaviour of $\log L$ as a function of $\log X$ has been found, so that any distinct phases can be identified along with their Bayes factors, as well as the overall evidence Z .

18 Statistics example

Let the coordinates θ have uniform prior over the 20-dimensional unit cube $[-\frac{1}{2}, \frac{1}{2}]^{20}$, and let the likelihood be

$$L(\theta) = 100 \prod_{i=1}^{20} \frac{1}{\sqrt{2\pi}u} \exp\left(-\frac{\theta_i^2}{2u^2}\right) + \prod_{i=1}^{20} \frac{1}{\sqrt{2\pi}v} \exp\left(-\frac{\theta_i^2}{2v^2}\right) \quad (48)$$

with $u = 0.01$ and $v = 0.1$. This represents a Gaussian “spike” of width 0.01 superposed on a Gaussian “plateau” of width 0.1. The Bayes factor favouring the spike is 100, and the evidence is $Z = 101$. There is only a single maximum, at the origin; this should surely be an easy problem. Yet $L(X)$ is partly convex (Figure 13), and an annealing program needs roughly a billion (e^{20}) trials to find the spike, and several times e^{25} to equilibrate properly. On the other hand, H is only 63.2, so nested sampling could reach and cross the spike and cover the whole range of Figure 13 in a mere 100 iterates.

Admittedly, about $N = 16$ points would be needed if the uncertainty from

$$\log Z \approx \log(101) \pm \sqrt{63.2/N} \quad (49)$$

$NH \approx 700N$ 迭代。

17 数据分析示例

并非只有具有棘手似然函数的大型问题才会表现出相变。以数据分析为例，考虑一个小型实验，用于测量单个坐标 θ ，其先验 $\pi(\theta)$ 在 $(0,1)$ 上是 平的。其数据 D 得出似然函数

$$L(\theta) = 0.99 q^{-1} e^{-\theta/q} + 0.01 \quad \text{with, say, } q = 10^{-9}. \quad (47)$$

这已经是一个递减函数，而嵌套采样的排序操作只是恒等操作， $X = \theta$ 。

对 (47) 的解释是，该实验预计以 99% 的可靠性进行。如果成功，似然 $L = q^{-1} \exp(-\theta/q)$ 将是合适的，这意味着 $\theta \approx 10^{-9}$ 被测量。如果失败，预计有 1% 的概率，似然将是无信息的 $L = 1$ ，因为设备只会返回一个随机结果。在退火过程中，原始的炽热相是失败模式。一个仅限于 $\beta \leq 1$ 的退火集合最不可能 找到 “工作” 模式，除非允许它进行数百万次试验，并且会错误地建议 “失败”，有 $Z = 0.01$ 。只有当 β 大幅增加到远超 1，达到 e^{1000} 以上时，该集合才可能在少于数百万次试验中找到工作模式。即便如此，样本也会向内崩溃，必须通过那些千个数量级退火回外。而且证据值会丢失。

对于嵌套采样，它持续跟踪 $\log X$ 而不是试图使用斜率，这些问题就很容易。所需的一切就是决心继续进行大约 NH 次收缩步骤，以到达并跨越主导模式，使用一组 N 点。到那时， $\log L$ 作为 $\log X$ 函数的行为已被找到，因此任何不同的阶段都可以被识别出来，连同它们的贝叶斯因子，以及整体证据 Z 。

18 统计学示例

设坐标 θ 在 20 维单位立方体 $[-\frac{1}{2}, \frac{1}{2}]^{20}$ 上具有均匀先验分布，并设似然函数为

$$L(\theta) = 100 \prod_{i=1}^{20} \frac{1}{\sqrt{2\pi}u} \exp\left(-\frac{\theta_i^2}{2u^2}\right) + \prod_{i=1}^{20} \frac{1}{\sqrt{2\pi}v} \exp\left(-\frac{\theta_i^2}{2v^2}\right) \quad (48)$$

其中包含 $u = 0.01$ 和 $v = 0.1$ 。这代表一个宽度为 0.01 的高斯 “尖峰” 叠加在一个宽度为 0.1 的高斯 “平台” 上。尖峰的贝叶斯因子为 100，证据为 $Z = 101$ 。只有一个最大值，在原点；这应该是一个很容易的问题。然而 $L(X)$ 部分是凸的（图 13），退火程序需要大约 10 亿 (e^{20}) 次试验才能找到尖峰，并且需要大约 e^{25} 次才能正确平衡。另一方面， H 只有 63.2，因此嵌套采样可以在 100 次迭代内到达并跨越尖峰，并覆盖图 13 的整个范围。

诚然，如果来自的不确定性，大约需要 $N = 16$ 点

$$\log Z \approx \log(101) \pm \sqrt{63.2/N} \quad (49)$$

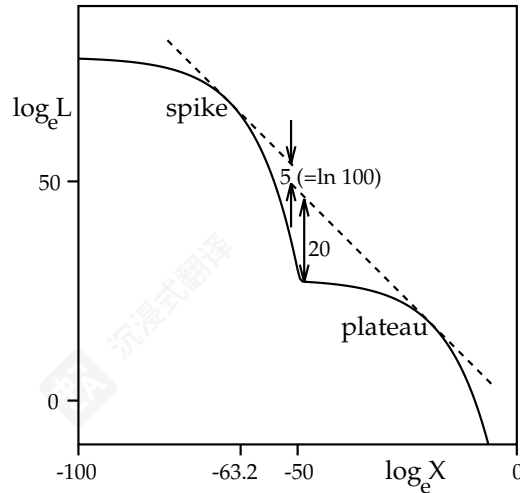


Figure 13: Gaussian spike on plateau. The spike is favoured by a Bayes factor $\exp(5)$, but annealing needs $\exp(20)$ trials to find it.

(and hence in the spike/plateau Bayes-factor logarithm) needs to be reduced to the ± 2 or so required to identify the favoured (spike) mode with reasonable confidence. That multiplies the computational load to 1600 evaluations, which remains less than a billion.

On the other hand, if the spike was moved off-centre to $(0.2, 0.2, 0.2, \dots)$, with likelihood

$$L(\theta) = 100 \prod_{i=1}^{20} \frac{1}{\sqrt{2\pi}u} \exp\left(-\frac{(\theta_i - 0.2)^2}{2u^2}\right) + \prod_{i=1}^{20} \frac{1}{\sqrt{2\pi}v} \exp\left(-\frac{\theta_i^2}{2v^2}\right), \quad (50)$$

then nested sampling too would be in difficulty. There are now two maxima over θ and, at the separatrix contour above which the phases separate, the aperture of the plateau is e^{35} times greater than that of the spike. This means that some huge number of trial points is needed to have a good chance of finding the spike, even though the $\log L / \log X$ plot is indistinguishable from Figure 13. That's impossible in practice. General multi-modality remains difficult.

19 Limiting cases

Two potentially awkward situations merit comment. The first (Figure 14a) is when $\log L$, as a function of $\log X$, has a discontinuity, visualised as a vertical cliff in the plot. Nested sampling is unaffected by this. The likelihood values that accompany a set of nested contours may change the termination condition, but they do not otherwise alter the algorithm's progress. The cliff doesn't matter. For the possible alternative method of multicanonical simulation (Berg and Neuhaus (1991)), the cliff would matter. Multicanonical simulation incorporates procedures to re-weight points artificially until $n(E) \equiv g = -dX/d \log L$ becomes nearly uniform. Having done that, the results can simulate any temperature. In this, the method is similar to nested sampling, though it lacks such clean numerical uncertainties. However, the discontinuous cliff will be difficult because the formal aim of multicanonical sampling becomes unattainable. There is a range of L over which there are no prior states at all, so their density

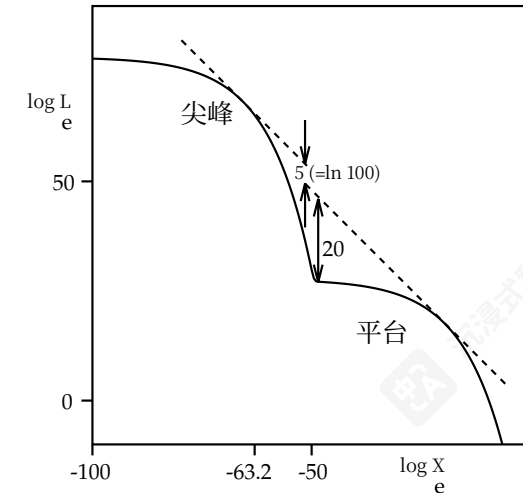


图13: 平台上的高斯尖峰。该尖峰受贝叶斯因子 $\exp(5)$ 的青睐, 但退火需要 $\exp(20)$ 次试验才能发现它。

这会使计算负载增加到1600次评估, 这仍然不到十亿。

另一方面, 如果将尖峰从中心移到 $(0.2, 0.2, 0.2, \dots)$, 且似然

$$L(\theta) = 100 \prod_{i=1}^{20} \frac{1}{\sqrt{2\pi}u} \exp\left(-\frac{(\theta_i - 0.2)^2}{2u^2}\right) + \prod_{i=1}^{20} \frac{1}{\sqrt{2\pi}v} \exp\left(-\frac{\theta_i^2}{2v^2}\right), \quad (50)$$

那么嵌套抽样也会陷入困境。现在有两个最大值在 θ 上, 在相位分离的分离线轮廓之上, 平台的孔径是尖峰的 e^{35} 倍。这意味着需要大量的试验点才能有较好的机会发现尖峰, 即使对数 L /对数 X 图与图13无法区分。这在实践中是不可能的。通用多模态仍然困难。

19 极限情况

两种潜在尴尬的情况值得评论。第一种(图14a)是当对数 L 作为对数 X 的函数出现不连续性时, 在图中表现为垂直的 cliff。嵌套采样不受此影响。伴随一组嵌套等高线的似然值可能会改变终止条件, 但不会改变算法的进展。cliff 并不重要。对于多尺度模拟的可能的替代方法 (Berg和 Neuhaus (1991)), cliff 就会很重要。多尺度模拟包含重新加权点的程序, 直到 $n(E) \equiv g = -dX/d \log L$ 变得几乎均匀。完成这些后, 结果可以模拟任何温度。在这方面, 该方法与嵌套采样相似, 尽管它缺乏如此清晰的数值不确定性。然而, 不连续的 cliff 将很难, 因为多尺度采样的正式目标变得无法实现。存在一个范围 L , 其中完全没有先验状态, 因此它们的密度

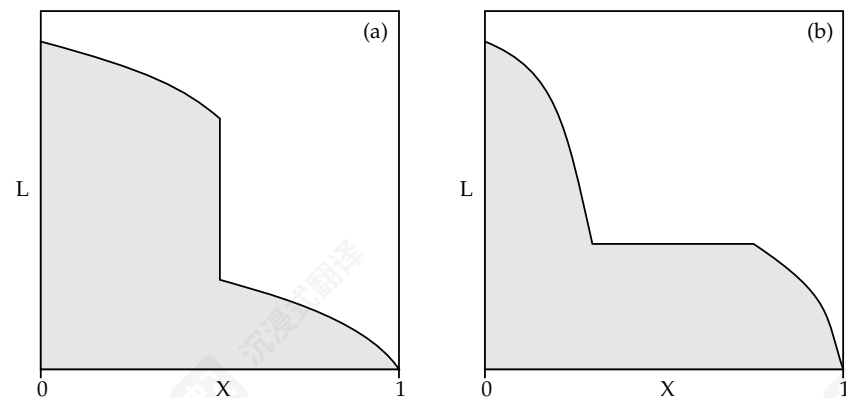


Figure 14: Cliff (a) and plateau (b) likelihood functions.

$n(\log L)$ can't be re-weighted to the desired constant. Nested sampling's $\log X$ is the abscissa of choice, not multicanonical simulation's $\log L$.

The other potentially awkward situation (Figure 14b) is when $\log L$, as a function of $\log X$, has a plateau. In this case, a finite prior mass is associated with one particular likelihood value. If the plateau covers a wide range, it may be difficult to locate the small interior domain in which L takes larger, possibly dominant, values. After all, the likelihood is offering no guidance, and the domain might have to be found (or not) by random exploration. It might then be very difficult to obtain useful new samples. Even so, it may be possible to generate them efficiently, by choosing labels ℓ so that small values anticipate easy transition to larger likelihoods. In this way, a judicious choice of extended likelihood L^+ (equation (6)) might give guidance even where L does not. Nested sampling would then continue to work without its samples becoming unduly expensive. However, “difficult” shapes, where the posterior breaks into separated islands whose mass is poorly predicted by their volume, will likely remain difficult.

20 Overview

Nested sampling reverses the accepted approach to Bayesian computation by putting the evidence first. A conventional collection of posterior samples can be computed as the calculation proceeds, but that is an optional extra, and quantified properties are better accumulated directly. The procedure runs with an evolving collection of N points, where N can be chosen small for speed or large for accuracy. In a specific application, it is the user's task to sample according to the prior density subject to a hard constraint on likelihood value. Helpfully, $N-1$ such points are already available as guides.

Nested sampling proceeds by systematically constraining the available prior mass, steadily decreasing its logarithm according to the shape of the likelihood contours. Its evolution path is independent of the actual likelihood values. This invariance over monotonic re-labelling may make it easier to find analytic properties that might imply almost-certain convergence. Currently, lack of convergence proofs for MCMC procedures is the main missing ingredient in Bayesian calculations, and with nested sampling attention can be focussed cleanly on the shape of the contours, without any distraction from likelihood values.

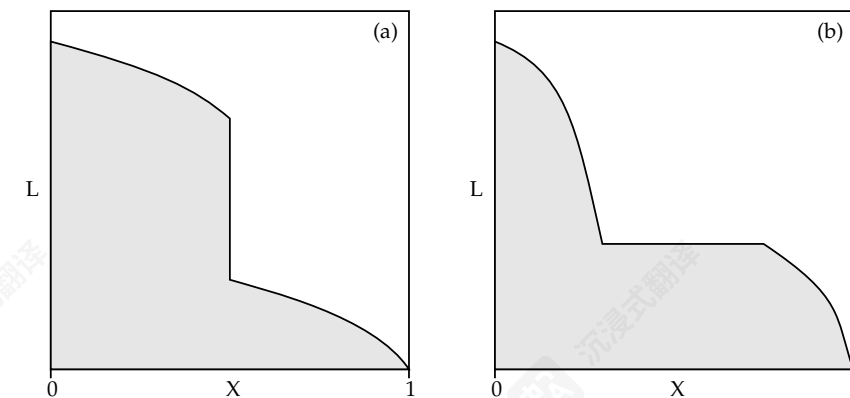


图14：Cliff (a) 和平台 (b) 似然函数。

$n(\text{对数 } L)$ 不能重新加权到所需的常数。嵌套采样的对数 X 是首选的横坐标，而不是多尺度模拟的对数 L 。

另一种潜在尴尬的情况（图14b）是当 $\log L$ 作为 $\log X$ 的函数出现平台时。在这种情况下，一个有限的先验质量与特定的似然值相关联。如果平台覆盖范围很广，可能很难定位 L 取较大值（可能占主导地位）的小内部域。毕竟，似然不提供任何指导，域可能必须通过随机探索来找到（或不找到）。然后可能很难获得有用的新样本。即便如此，也可能通过选择标签 ℓ 使小值能轻松过渡到较大的似然值来高效生成它们。通过这种方式，一个明智的扩展似然 L^+ （方程(6)）即使在 L 不起作用的地方也能提供指导。嵌套采样将继续工作，而其样本不会变得过贵。然而，\困难\形状，其中后验分解为分离的岛屿，其质量与其体积的预测不佳，很可能仍然很困难。

20 概述

嵌套采样通过将证据 $\mathfrak{f}$ 放在首位，颠覆了贝叶斯计算的公认方法。在计算过程中，可以计算常规的后验样本集合，但这只是一个可选的附加功能，而量化 $\mathfrak{f}$ 的属性最好直接累积。该过程使用一个不断发展的 N 点集合运行，其中 N 可以选择得小以加快速度或选择得大以提高精度。在一个特定的应用中，用户的任务是根据先验密度进行采样，同时受到对似然值的硬约束。幸运的是， $N-1$ 这样的点已经可用作为指导。

嵌套抽样通过系统地约束可用先验质量，根据似然等高线的形状逐步减小其对数。其演化路径独立于实际的似然值。这种单调重标记下的不变性可能使寻找可能暗示几乎必然收敛的解析特性变得更容易。目前，MCMC程序缺乏收敛证明是贝叶斯计算中的主要缺失环节，而嵌套抽样可以清晰地关注等高线的形状，而不会受到似然值的任何干扰。

We do not address the errors that would arise from imperfect sampling of the box within a given likelihood contour. The uncertainties that arise from the method itself, though, are understood and controllable. Numerical uncertainties accompany estimates of evidence and any quantified property. These are properly probabilistic, and not derived from jack-knife or similar frequentist fixup. In short, nested sampling follows the rules of probability calculus, so is Bayesian in nature, as befits an algorithm for Bayesian computation. As usual with probabilistic procedures, uncertainties decrease as $\sqrt{N}$ whereas the computational load is proportional to N .

For some problems, it may be possible to find provably exact samples (Propp and Wilson (1996)) within a likelihood contour, and thus remove any doubt concerning imperfect sampling. Because the probabilities p_i of the nested samples are calculated essentially perfectly, the posterior samples generated through nested sampling would also be exact, as would the quantification statistics Q derived from the complete nested sequence. If exact samples turn out to be easier to find within a likelihood contour than with respect to the full posterior, this would extend the currently small class of problems amenable to exact sampling, with the added benefit of obtaining the evidence value.

Nested sampling has some similarity with annealing in that it works by compressing the available domain. The hard outer constraint on likelihood happens to give a similar restriction on step-length to that applying in standard Metropolis-type detailed balance, so the new method should offer no great gain or loss of computational speed, as compared with annealing under an efficient schedule. Even so, nested sampling is more fundamental than annealing, in that it gives a direct view of the underlying density of states $g^*(L)$ as it steps steadily inward. More importantly, it can deal straightforwardly with convex likelihood functions that exhibit first-order phase changes. Nested sampling has the simplicity and generality that speak of wide-ranging power.

21 Acknowledgments

This work was supported by Maximum Entropy Data Consultants Ltd of Cambridge (England). It was Ken Hanson at the 23rd maximum entropy workshop who re-awakened my long-standing interest in probabilistic integration. I also wish to thank David MacKay and the Inference group journal club at Cambridge for useful encouragement and advice, and for hosting nested-sampling papers and programs (currently at www.inference.phy.cam.ac.uk/bayesys). An early account of the method was given in Skilling (2004). This presentation has been considerably improved by the JBA referees, who well succeeded in their task of being critical.

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Duane, S., A. D. Kennedy, B. J. Pendleton, and D. Roweth. 1987. Hybrid Monte Carlo. *Phys. Lett. B* 195: 216–222. 838

Gelman, A. and X.-L. Meng. 1998. Simulating normalizing constants: from importance sampling to bridge sampling to path sampling. *Statistical Science* 13: 163–185. 834

我们不讨论从给定似然等高线内不完美抽样盒子中产生的误差。然而，方法本身产生的误差是可理解和可控的。证据和任何量化特性的估计都伴随着数值不确定性。这些是真正的概率不确定性，而不是从刀刀法或类似的频率 fixup推导出来的。简而言之，嵌套抽样遵循概率演算规则，因此具有贝叶斯性质，作为贝叶斯计算的算法。与概率程序通常一样，不确定性随着 $\sqrt{N}$ 而减少，而计算负载与 N 成正比。

对于某些问题，可能可以在似然轮廓内找到可证明精确的样本（Propp和Wilson（1996）），从而消除对不完美抽样的任何疑虑。由于嵌套样本的概率 p_i 基本上被完美地计算，通过嵌套抽样生成的后验样本也将是精确的，从完整的嵌套序列中推导出的量化统计量 Q 也将是精确的。如果精确样本在似然轮廓内比相对于完整后验分布更容易找到，这将扩展目前很小的一类适合精确抽样的问题的类别，并额外获得证据值。

嵌套抽样与退火算法有相似之处，因为它通过压缩可用域来工作。对似然函数的硬外约束恰好给出了与标准Metropolis类型详细平衡中应用的步长限制相似的约束，因此新方法与在有效调度下的退火相比，计算速度应该没有很大的增益或损失。即便如此，嵌套抽样比退火更基本，因为它在稳步向内移动时，能直接看到潜在的状态密度 $g^*(L)$ 。更重要的是，它可以直接处理表现出 μ -一阶相变的凸似然函数。嵌套抽样具有简单性和普遍性，这表明它具有广泛的能力。

21 致谢

这项工作得到了剑桥（英国）最大熵数据咨询有限公司的支持。是第23届最大熵研讨会上的肯·汉森重新唤起了我对概率积分的长期兴趣。我还要感谢大卫·麦凯和剑桥推理小组期刊俱乐部提供的宝贵鼓励和建议，以及他们托管嵌套采样论文和程序（目前位于 www.inference.phy.cam.ac.uk/bayesys）。该方法早期的介绍见于斯金林（2004）的著作。本次演示得到了JBA审稿人的显著改进，他们出色地完成了其批判性任务。

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Appendix

```
/*=====
TOY NESTED SAMPLING PROGRAM IN ‘C’ by John Skilling, Aug 2005
GNU GENERAL PUBLIC LICENSE software
http://www.gnu.org/copyleft/gpl.html
=====*/
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
#include <float.h>
#define UNIFORM ((rand()+0.5) / (RAND_MAX+1.0)) // Uniform(0,1)
#define logZERO (-DBL_MAX * DBL_EPSILON) // log(0)
#define PLUS(x,y) (x>y ? x+log(1+exp(y-x)) : y+log(1+exp(x-y)))
// logarithmic addition log(exp(x)+exp(y))

/* YOU MUST PROGRAM THIS FROM HERE ++++++
typedef struct
{
    ANYTYPE theta; // YOUR coordinates
    double logL; // logLikelihood = ln Prob(data | theta)
    double logWt; // ln(Weight), summing to SUM(Wt) = Evidence Z
```

Hastings, W. K. 1970. Monte Carlo sampling methods using Markov chains and their applications. *Biometrika* 57: 97–109. [834](#)

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附录

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#define logZERO (-DBL_MAX * DBL_EPSILON) // log(0)
#define PLUS(x,y) (x>y ? x+log(1+exp(y-x)) : y+log(1+exp(x-y)))
// logarithmic addition log(exp(x)+exp(y))

/* YOU MUST PROGRAM THIS FROM HERE ++++++
typedef struct
{
    ANYTYPE theta; // YOUR coordinates
    double logL; // logLikelihood = ln Prob(data | theta)
    double logWt; // ln(Weight), summing to SUM(Wt) = Evidence Z
```

```

} Object;
double logLhood(ANYTYPE theta){...} // logLikelihood function
void Prior (Object* Obj){...} // Set Object according to prior
void Explore(Object* Obj, double logLstar){...}
// Evolve Object within likelihood constraint
----- UP TO HERE */

int main(void)
{
#define N 100 // # Objects
#define MAX 9999 // max # Samples (allow enough)
    Object Obj[N]; // Collection of N objects
    Object Samples[MAX]; // Objects defining posterior
    double logw; // ln(width in prior mass)
    double logLstar; // ln(Likelihood constraint)
    double H = 0.0; // Information, initially 0
    double logZ = logZERO; // ln(Evidence Z, initially 0)
    double logZnew; // Updated logZ
    int i; // Object counter
    int copy; // Duplicated object
    int worst; // Worst object
    int nest; // Nested sampling iteration count
    double end = 2.0; // Termination condition nest = end * N * H
// Set prior objects
    for( i = 0; i < N; i++ )
        Prior( &Obj[i] );
// Outermost interval of prior mass
    logw = log(1.0 - exp(-1.0 / N));
// Begin Nested Sampling loop ++++++
    for( nest = 0; nest <= end * N * H; nest++ )
    {
// Worst object in collection, with Weight = width * Likelihood
        worst = 0;
        for( i = 1; i < N; i++ )
            if( Obj[i].logL < Obj[worst].logL )
                worst = i;
        Obj[worst].logWt = logw + Obj[worst].logL;
// Update Evidence Z and Information H
        logZnew = PLUS(logZ, Obj[worst].logWt);
        H = exp(Obj[worst].logWt - logZnew) * Obj[worst].logL
            + exp(logZ - logZnew) * (H + logZ) - logZnew;
        logZ = logZnew;
// Posterior Samples (optional, care with storage overflow)
        Samples[nest] = Obj[worst];
// Kill worst object in favour of copy of different survivor
        do copy = (int)(N * UNIFORM) % N; // force 0 <= copy < N

```

```

} Object;
double logLhood(ANYTYPE theta){...} // logLikelihood function
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void Explore(Object* Obj, double logLstar){...}
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        Prior( &Obj[i] );
// Outermost interval of prior mass
    logw = log(1.0 - exp(-1.0 / N));
// Begin Nested Sampling loop ++++++
    for( nest = 0; nest <= end * N * H; nest++ )
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        worst = 0;
        for( i = 1; i < N; i++ )
            if( Obj[i].logL < Obj[worst].logL )
                worst = i;
        Obj[worst].logWt = logw + Obj[worst].logL;
// Update Evidence Z and Information H
        logZnew = PLUS(logZ, Obj[worst].logWt);
        H = exp(Obj[worst].logWt - logZnew) * Obj[worst].logL
            + exp(logZ - logZnew) * (H + logZ) - logZnew;
        logZ = logZnew;
// Posterior Samples (optional, care with storage overflow)
        Samples[nest] = Obj[worst];
// Kill worst object in favour of copy of different survivor
        do copy = (int)(N * UNIFORM) % N; // force 0 <= copy < N

```



```

        while( copy == worst && N > 1 );    // don't kill if N=1
        logLstar = Obj[worst].logL;    // new likelihood constraint
        Obj[worst] = Obj[copy];        // overwrite worst object
// Evolve copied object within constraint
        Explore( &Obj[worst], logLstar );
// Shrink interval
        logw -= 1.0 / N;
    } // ----- end nested sampling loop
// Begin optional final correction, should be small ++++++
    logw = -(double)nest / (double)N - log((double)N);    // width
    for( i = 0; i < N; i++ )
    {
        Obj[i].logWt = logw + Obj[i].logL;    // width * Likelihood
// Update Evidence Z and Information H
        logZnew = PLUS(logZ, Obj[i].logWt);
        H = exp(Obj[i].logWt - logZnew) * Obj[i].logL
            + exp(logZ - logZnew) * (H + logZ) - logZnew;
        logZ = logZnew;
// Posterior Samples (optional, care with storage overflow)
        Samples[nest++] = Obj[i];
    } // ----- end optional final correction
// Exit with evidence Z, information H, and posterior Samples
    printf("#samples = %d\n", nest);
    printf("Evidence:  ln(Z) = %g +- %g\n", logZ, sqrt(H/N));
    printf("Information:  H = %g nats = %g bits\n", H, H/log(2.));
// You can now accumulate results from Samples[0...nest-1]
    return 0;
}

```

```

        while( copy == worst && N > 1 );    // don't kill if N=1
        logLstar = Obj[worst].logL;    // new likelihood constraint
        Obj[worst] = Obj[copy];        // overwrite worst object
// Evolve copied object within constraint
        Explore( &Obj[worst], logLstar );
// Shrink interval
        logw -= 1.0 / N;
    } // ----- end nested sampling loop
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    logw = -(double)nest / (double)N - log((double)N);    // width
    for( i = 0; i < N; i++ )
    {
        Obj[i].logWt = logw + Obj[i].logL;    // width * Likelihood
// Update Evidence Z and Information H
        logZnew = PLUS(logZ, Obj[i].logWt);
        H = exp(Obj[i].logWt - logZnew) * Obj[i].logL
            + exp(logZ - logZnew) * (H + logZ) - logZnew;
        logZ = logZnew;
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        Samples[nest++] = Obj[i];
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// You can now accumulate results from Samples[0...nest-1]
    return 0;
}

```

